

CHM 3120: Introduction to Analytical Chemistry

Unit 5 - Section 2:

Potentiometry

Chapters 19

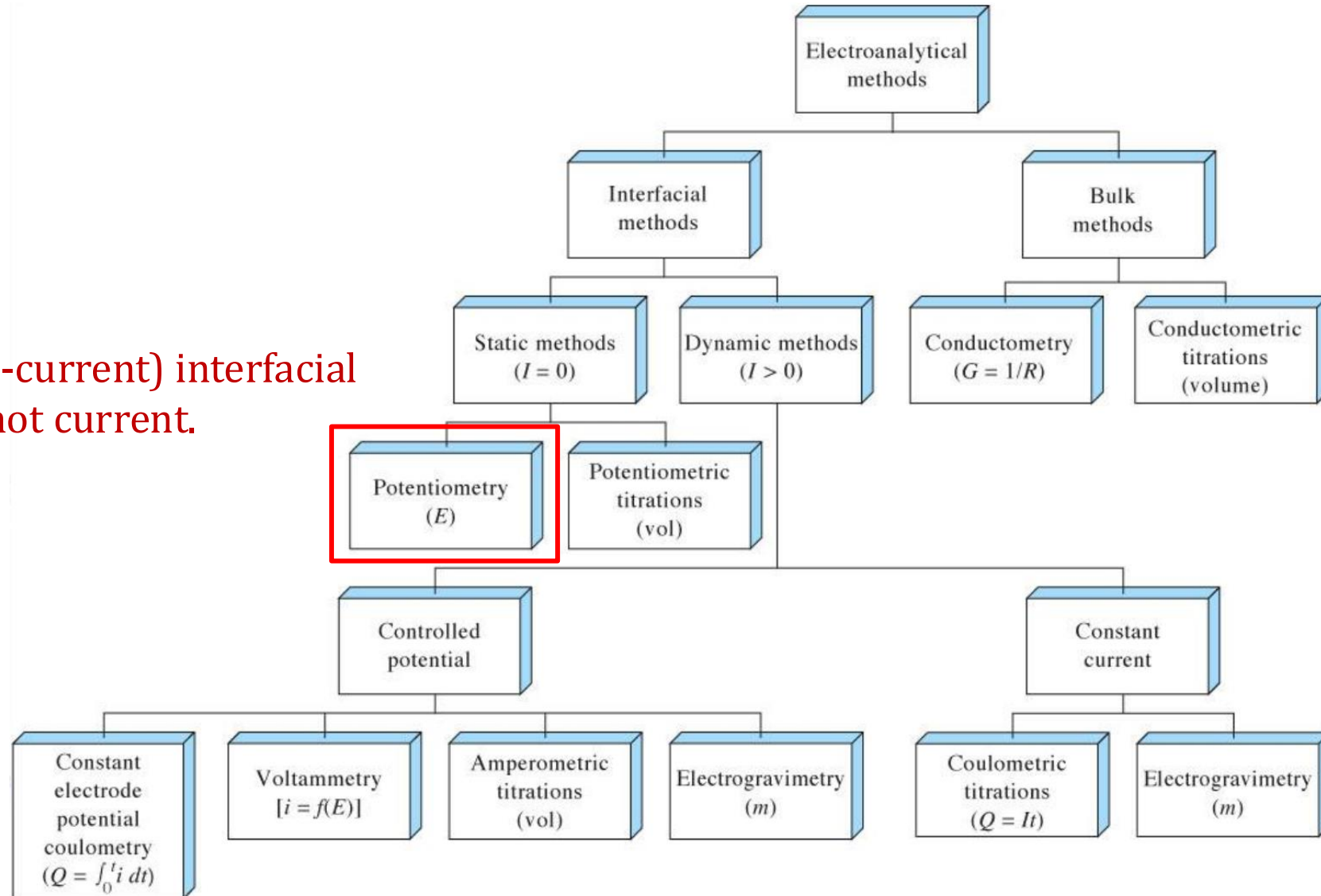
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Electroanalytical Methods: Classification

Potentiometry belongs to static (no-current) interfacial methods → we measure potential, not current.



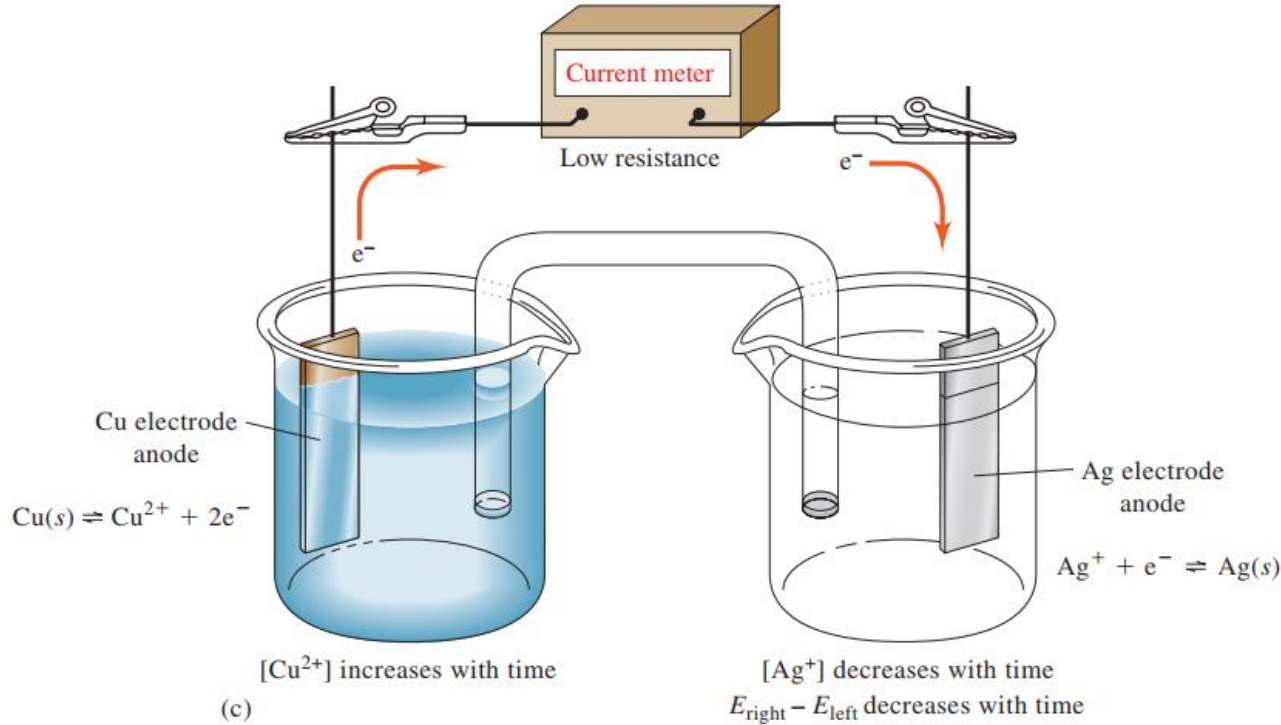
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What Is Potentiometry?

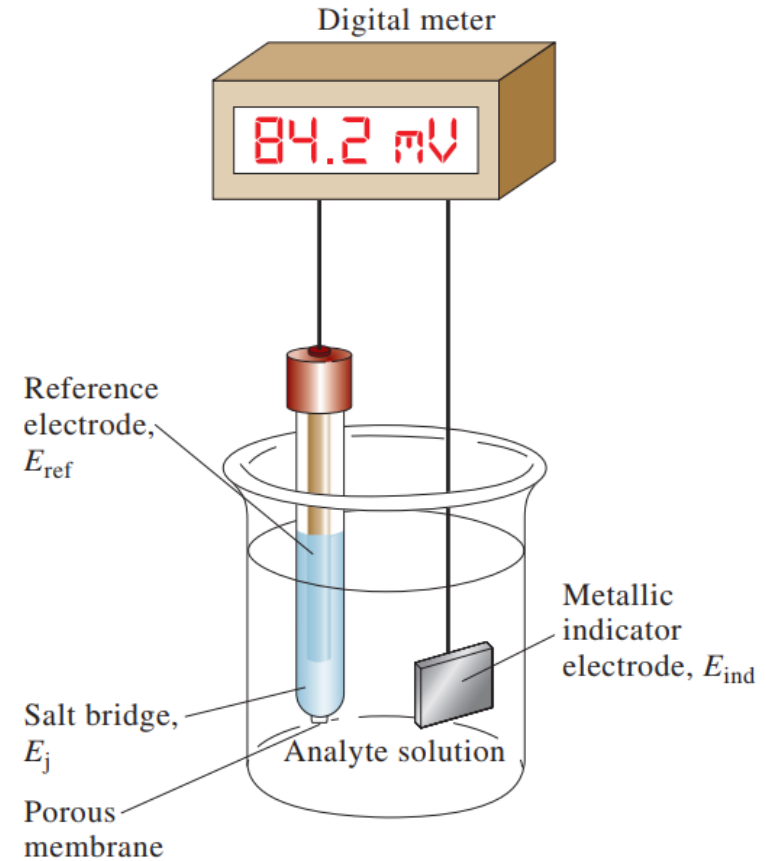
- Potentiometry is a static (no-current) electrochemical technique.
- It measures the potential difference between two electrodes in an electrochemical cell under open-circuit conditions (no current flow).
- The measured potential is related to the **activity of ions** in solution through the Nernst equation.
- The pH meter is the most common example of a potentiometric instrument, based on a glass-membrane electrode selective to H^+ activity.
- Potentiometric methods are simple, reliable, and widely used for **ion-selective measurements** in analytical chemistry.
- Potential varies logarithmically with ion activity (Nernst relationship).



Measuring Potentials: From Galvanic to Potentiometric Cells



In a galvanic cell: electrons flow — current is measured (dynamic system).

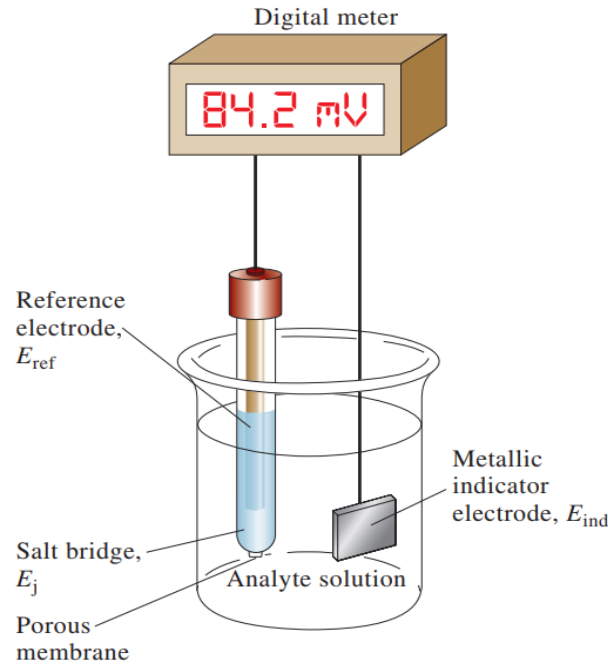


In potentiometry: no current flows — only the potential difference is measured.

Components and Cell Potential Equation

Every potentiometric measurement consists of three potential contributions:

By convention, the **reference electrode** is always treated as the left-hand electrode in potentiometric measurements.



The **indicator (working) electrode** is immersed in the analyte solution and develops a potential that depends on the activity of the target ion.

The **salt bridge (or junction)** connects the two half-cells, allowing charge balance while minimizing ion mixing. It introduces a small liquid junction potential that cannot be fully eliminated.

Components and Cell Potential Equation



$$E_{cell} = E_{ind} - E_{ref} - E_j$$

- The measured cell potential (E_{cell}) is the difference between the indicator and reference electrode potentials, corrected for the small liquid junction potential.
- E_j is often small enough to be neglected, but it may cause minor systematic errors.
- The magnitude of E_{cell} depends primarily on the activity of ions to which the indicator electrode is selective.

1. Reference Electrode

- The reference electrode provides a **stable and well-defined potential** against which the indicator electrode potential is measured.
- It can be a **standard hydrogen electrode (SHE)**, but this is rarely used because it is difficult to maintain and impractical for routine measurements.
- An ideal reference electrode should have a potential that is:
 - **Accurately** known and highly reproducible.
 - **Constant** over time and independent of the analyte composition.
 - **Insensitive** to small current flow and environmental changes.
- It should be also rugged, easy to assemble, and maintain a constant potential while passing minimal current.

Common examples:

- **Calomel electrode (Hg/Hg_2Cl_2 , KCl)**
- **Silver/Silver chloride electrode ($Ag/AgCl$, KCl)**

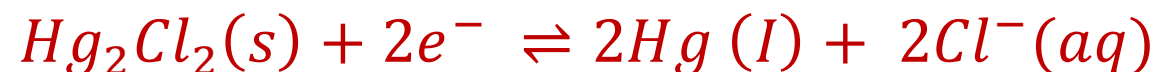
Calomel Reference Electrodes

The calomel electrode consists of liquid mercury (Hg) in contact with mercury(I) chloride (Hg_2Cl_2 : calomel) and a KCl solution of known concentration.

Cell notation: $Hg | Hg_2Cl_2(sat'd), KCl(x\ M) ||$

where x indicates the molarity of KCl (commonly 0.1 M, 1.0 M, or saturated).

The half-cell reaction is:

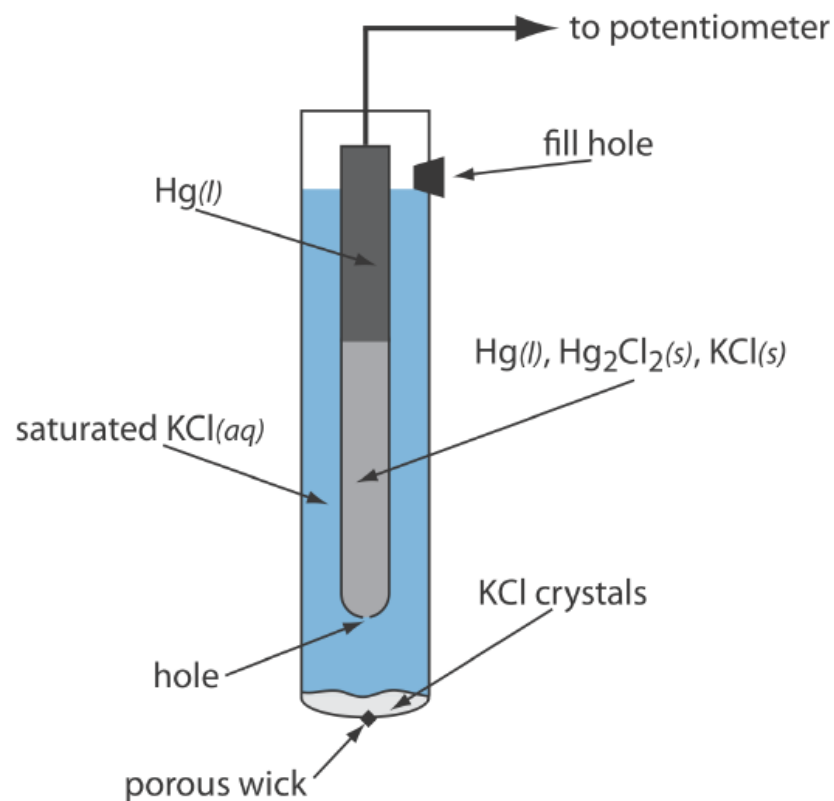


$$E = E_{Hg_2Cl_2/Hg}^\circ - \frac{0.059}{2} \log[(a_{Cl^-})^2]$$

For saturated KCl (25 °C):

$$E = 0.244\ V - \frac{0.059}{2} \log[(a_{Cl^-})^2]$$

a = activity of Cl^- ($\approx$ concentration at low ionic strength; will be discussed shortly)



Calomel Reference Electrodes

$$E = E_{Hg_2Cl_2/Hg}^{\circ} - \frac{0.059}{2} \log[(a_{Cl^-})^2] \xrightarrow{\log[(a_{Cl^-})^2] = 2 \log(a_{Cl^-})} E = E_{Hg_2Cl_2/Hg}^{\circ} - \frac{0.059}{2} \times 2 \log(a_{Cl^-})$$
$$\longrightarrow E = E_{Hg_2Cl_2/Hg}^{\circ} - 0.059 \log(a_{Cl^-})$$

The electrode **potential depends on a_{Cl^-} ($= [Cl^-]$)**, so the KCl concentration must always be specified.

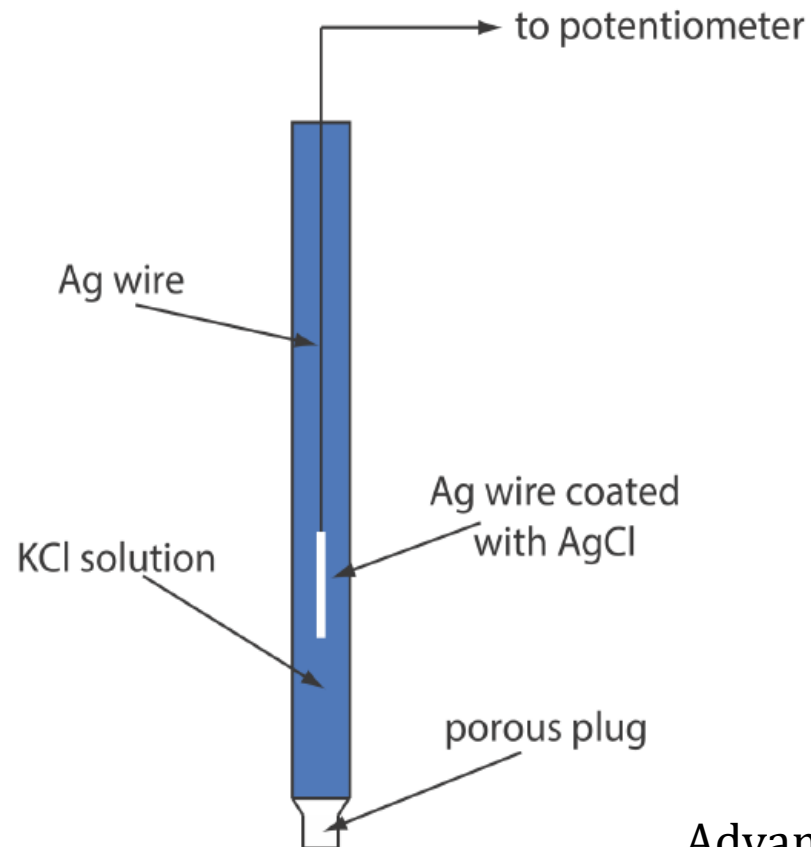
Typical potentials (vs SHE, at 25 °C):

Saturated KCl (SCE)	→ +0.2444 V
3.5 M KCl	→ +0.250 V
0.1 M KCl	→ +0.3356 V

Often we use Saturated Calomel Electrode (SCE):



Silver/Silver Chloride Electrodes



- The Ag/AgCl reference electrode consists of a silver wire coated with silver chloride (AgCl) and immersed in a KCl solution of known concentration.
- The KCl solution is usually **saturated or 3.5 M**, ensuring stable and reproducible potential.
- Cell notation: $Ag | AgCl(sat'd), KCl(sat'd) ||$

The half-cell reaction is: $AgCl(s) + e^- \rightleftharpoons Ag(s) + Cl^-(aq)$

Nernst equation (25 °C): $E = E_{AgCl/Ag}^\circ - 0.059 \log(a_{Cl^-})$

For saturated KCl (25 °C): $E = 0.199 V - 0.059 \log(a_{Cl^-})$

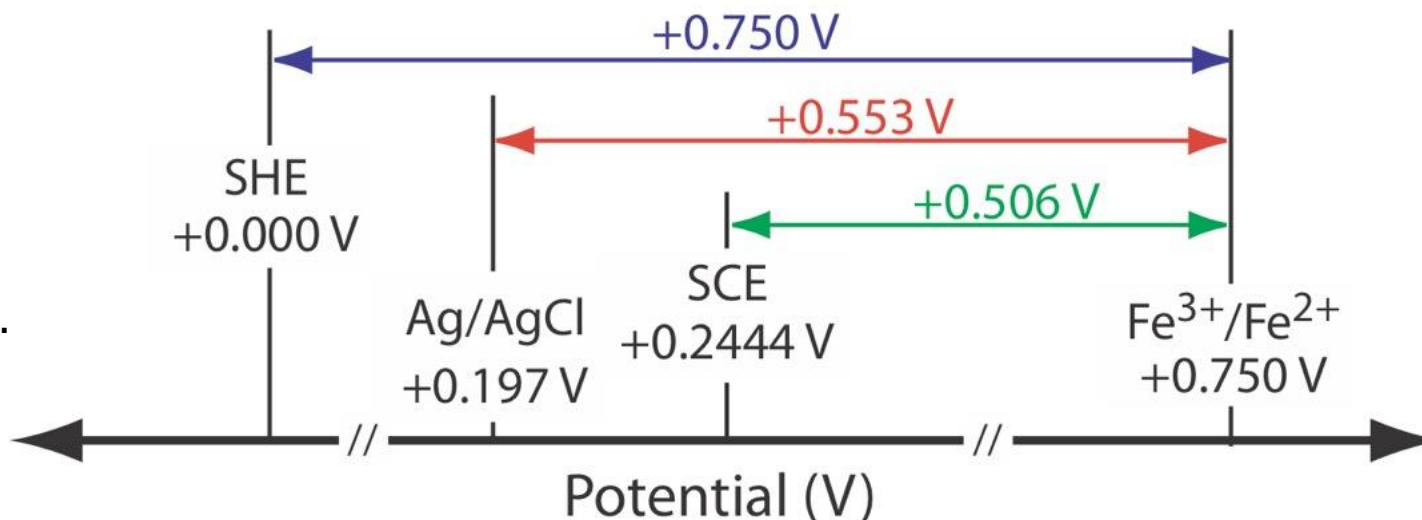
Advantages over Calomel Electrode:

- Ag/AgCl usable > 60 °C; Calomel unstable above this temperature.
- Calomel (Hg) less reactive; Ag^+ can bind proteins → junction clogging.
- Provides reproducible potential and is easy to maintain.

Reference Electrode Potentials (vs SHE)

Temperature (°C)	0.1 M Calomel	3.5 M Calomel	Sat'd Calomel	3.5 M Ag/AgCl	Sat'd Ag/AgCl
15	0.3362	0.254	0.2511	0.212	0.209
20	0.3359	0.252	0.2479	0.208	0.204
25	0.3356	0.250	0.2444	0.205	0.199
30	0.3351	0.248	0.2411	0.201	0.194
35	0.3344	0.246	0.2376	0.197	0.189

The diagram shows electrode potentials relative to SHE, highlighting how different reference electrodes shift measured voltages.



What Is Activity? Why Not Just Concentration?

Discussed in Chapter 8B Activity Coefficients

Activity accounts for ionic interactions: In real solutions, ions interact with each other, so their effective concentration (activity) is not equal to their molar concentration.

$$a_X = [X]\gamma_X$$

- At very low ionic strength: $\gamma \approx 1 \rightarrow \text{activity} \approx \text{concentration}$
- As ionic strength increases: effective concentration decreases $\rightarrow \gamma$ decreases
- Higher charge $\rightarrow$ larger deviation (e.g., Al^{3+} , $\text{Fe}(\text{CN})_6^{4-}$)

The **Debye-Hückel equation** relates the activity coefficient to ionic strength:

$$-\log \gamma_X = \frac{0.51 z_X^2 \sqrt{\mu}}{1 + 3.3 a_X \sqrt{\mu}}$$

for very low ionic strength ($\mu < 0.01 \text{ M}$): Denominator $\approx 1 \rightarrow$ limiting law:

$$-\log \gamma_X \approx 0.51 z_X^2 \sqrt{\mu}$$

- Activity coefficient γ decreases as ionic strength increases
- Effect is stronger for highly charged ions
- Neutral molecules: $\gamma \approx 1$ (almost no effect)

a_X : activity
 $[X]$: molar concentration
 γ_X : activity coefficient (dimensionless)
 z_X = charge of the ion
 μ = ionic strength
 a_X = effective hydrated ion diameter (nm)

Why Activity Matters in Electrochemistry

Nernst equation uses activity $E = E^\circ - \frac{0.0592}{n} \log a$ since $a = [X]\gamma$

Electrode potentials depend on both concentration AND ionic strength.

Reference electrodes rely on constant ionic strength

Example: Ag/AgCl $E = E_{AgCl/Ag}^\circ - 0.0592 \log a_{Cl^-}$

With saturated KCl: $[Cl^-]$ fixed $\rightarrow \gamma_{Cl^-}$ approximately constant $\rightarrow$ activity is constant, giving a stable reference potential.

Why E°' appears: E°' (formal potential) includes the effect of activity coefficients at a defined ionic strength

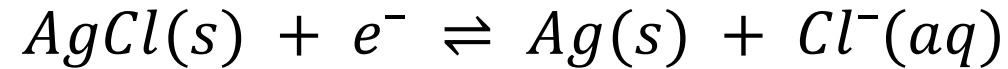
Formal potentials for Fe^{3+}/Fe^{2+} half-reaction in five different solvents.

Half reaction	E° (V)	E°' (V)
$Fe^{3+} + e^- \rightleftharpoons Fe^{2+}$	0.771	0.70 in 1 M HCl
		0.767 in 1 M $HClO_4$
		0.746 in 1 M HNO_3
		0.68 in 1 M H_2SO_4
		0.44 in 0.3 M H_3PO_4

Example: Activity of Cl^- in a Saturated KCl Ag/AgCl Electrode

We can determine the activity $a_{(\text{Cl}^-)}$ from the measured E of the saturated Ag/AgCl reference electrode.

Half-reaction:



Nernst equation:

$$E = E^\circ - 0.05916 \log(a_{(\text{Cl}^-)})$$

Known values at 25 °C:

Solve for activity: $0.197 = 0.222 - 0.059 \log(a_{(\text{Cl}^-)}) \rightarrow a_{(\text{Cl}^-)} = 2.65$

Compare to formal concentration: $a = \gamma \times c \rightarrow \gamma = 2.65/4.79 \rightarrow \gamma \approx 0.55$

Even though $[\text{Cl}^-]=4.79 \text{ M}$, it “behaves” like it’s only 2.6 M because $\gamma < 1$.

Example: Activity of Cl^- in a Saturated KCl Ag/AgCl Electrode

Extended Debye-Hückel Equation:

$$\log \gamma = \frac{-0.51z^2\sqrt{\mu}}{1+(\alpha\sqrt{\mu}/305)}$$

Known values for Cl^- in saturated KCl: $z = -1$; $\mu \approx 4.79$ ($\approx [\text{KCl}]$, 1:1 electrolyte); $\alpha \approx 300$ pm

$$\log \gamma = \frac{-0.51(1)^2\sqrt{4.79}}{1 + \left(\frac{300\sqrt{4.79}}{305}\right)} \rightarrow \gamma \approx 0.44$$

Debye-Hückel: $\gamma \approx 0.44$

vs.

From activity measurement: $\gamma \approx 0.55$

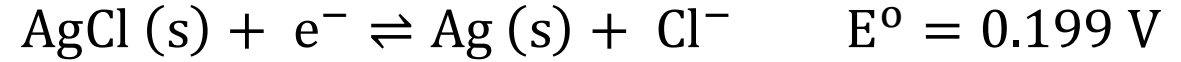
Debye-Hückel becomes inaccurate at high ionic strength ($\mu > 0.1$ M). Saturated KCl is very concentrated, so activity must be obtained experimentally.

What does all these mean?

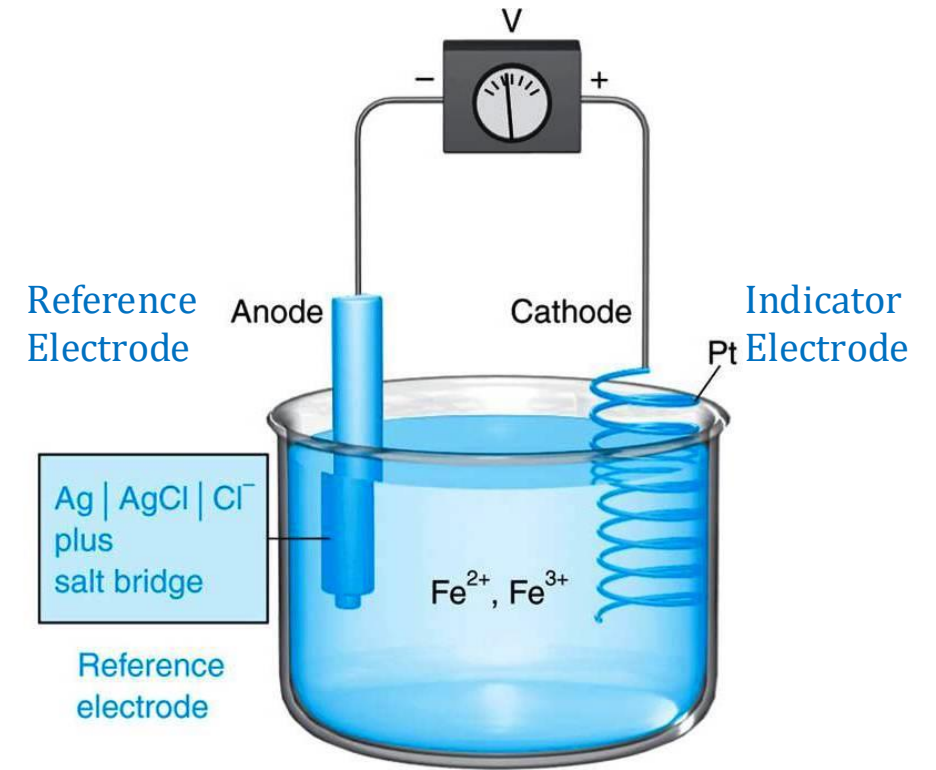
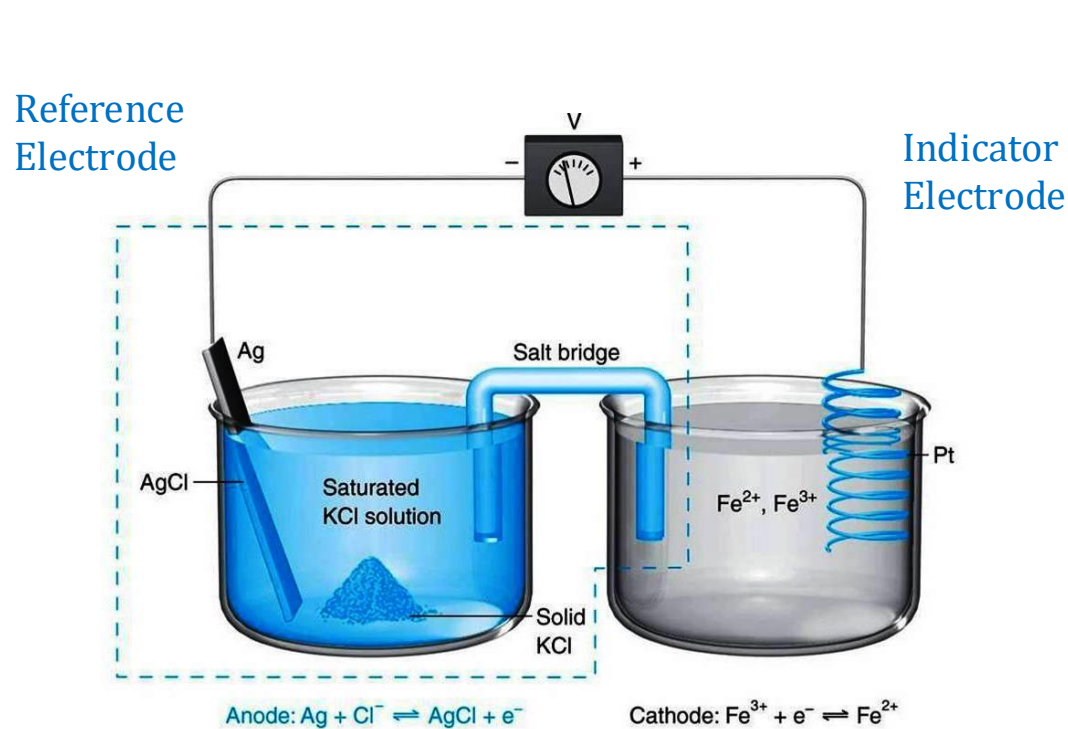
- Activity coefficients must be considered in electrochemistry, especially when dealing with reference electrodes like Ag/AgCl or calomel, where Cl^- concentration strongly affects the electrode potential.
- Saturated KCl fixes both $[\text{Cl}^-]$ and γ_{Cl^-} , giving a constant a_{Cl^-}
- A constant activity produces a stable and reproducible reference potential

Example: Ag/AgCl Reference + Fe³⁺/Fe²⁺ System

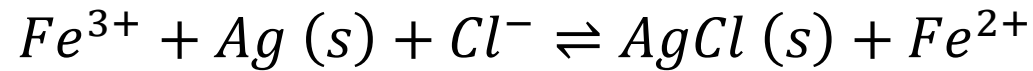
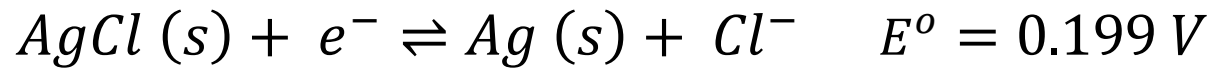
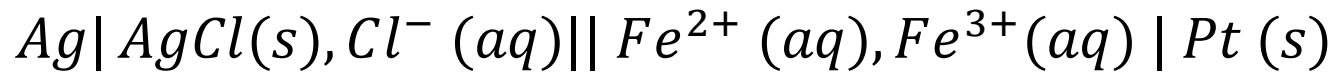
Reference electrode:



Indicator electrode (responds to analyte):



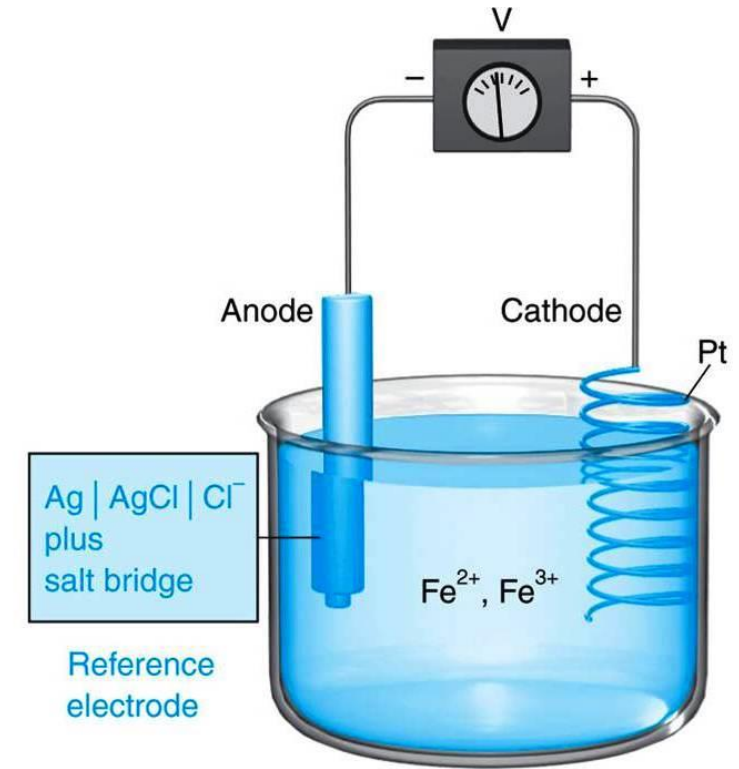
- The voltmeter measures the difference between these two electrode (half cell) potentials.
- The reference electrode provides a fixed, stable potential, while the indicator electrode responds to the $\text{Fe}^{3+}/\text{Fe}^{2+}$ ratio.



$$E_{cell} = E_{ind} - E_{ref} - E_j$$

$$E_{cell} = E_{cell}^{\circ} - \frac{0.0592 V}{n} \log Q$$

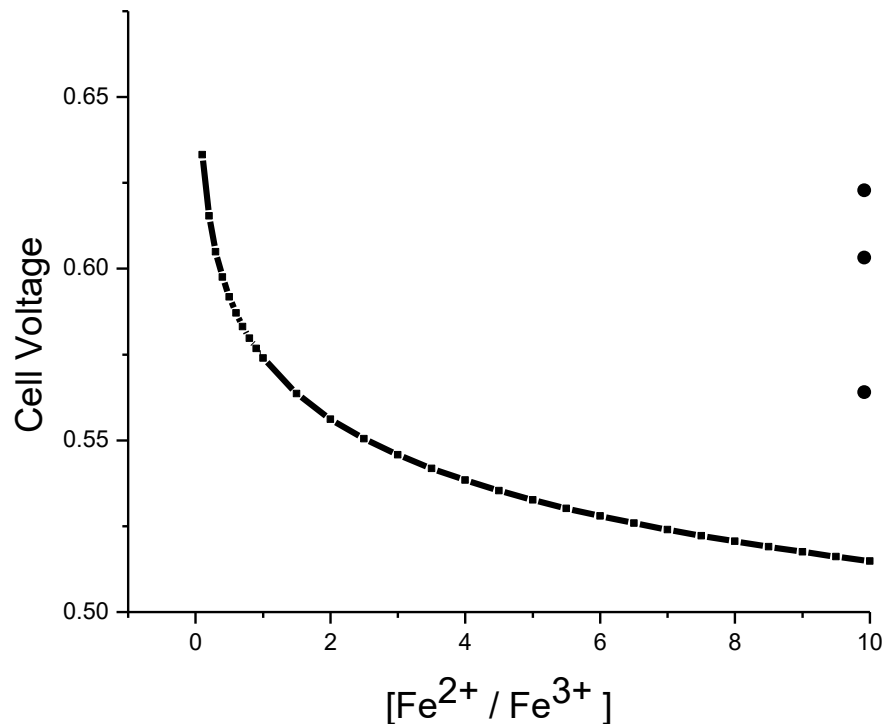
$$E = \left\{ 0.771 - 0.0592 \log \frac{[Fe^{2+}]}{[Fe^{3+}]} \right\} - \{ 0.199 - 0.059 \log [Cl^{-}] \}$$



$$E = \left\{ 0.771 - 0.0592 \log \left(\frac{[Fe^{2+}]}{[Fe^{3+}]} \right) \right\} - \{ 0.199 - 0.059 \log [Cl^{-}] \}$$

If $[Cl^{-}]$ is constant (saturated KCl):

$$E_{cell} = 0.572 V - 0.0592 \log Q$$



Q

- Cell voltage decreases as the ratio $Q = [Fe^{2+}] / [Fe^{3+}]$ increases.
- The reference potential is fixed; the indicator follows the Nernst equation.
- The curve shows the measurable dependence of cell potential on the $[Fe^{2+}] / [Fe^{3+}]$ ratio.

If we rewrite:

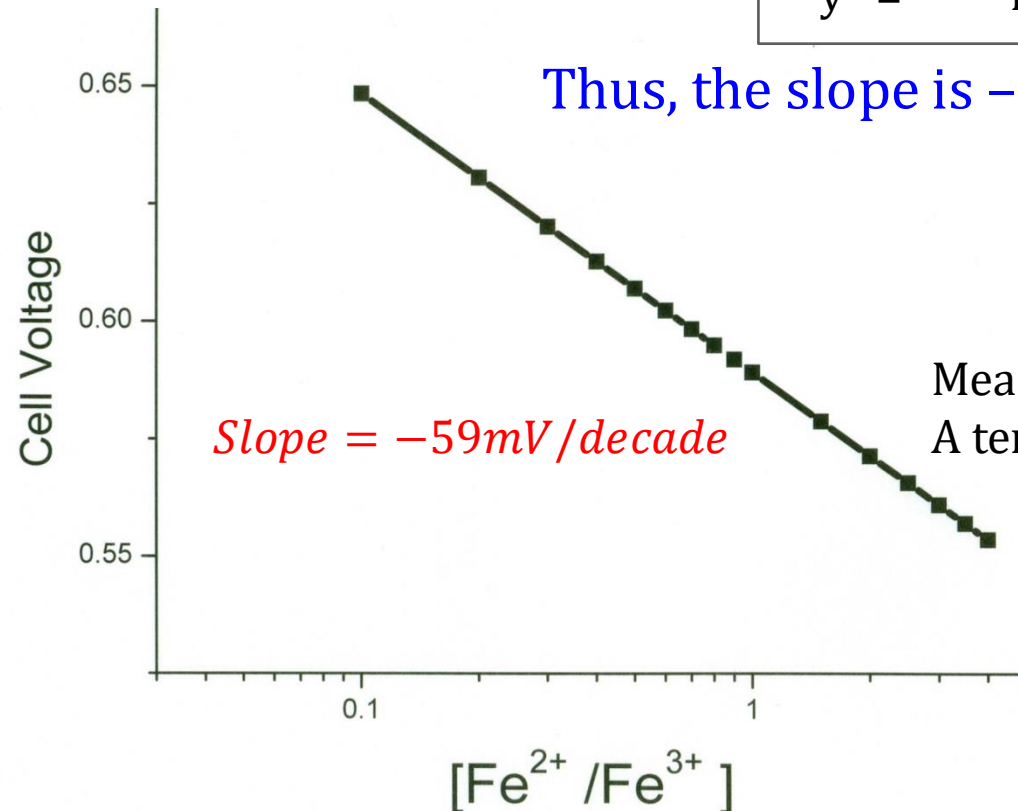
$$E_{cell} = 0.572 V - 0.0592 \log(Q)$$

As a linear expression: $E_{cell} = -0.0592 \log(Q) + 0.572$

If we create a calibration curve, we can use that to find the concentration of an unknown

$$y = m x + b$$

Thus, the slope is $-0.0592 V = -59 \text{ mV}$ per decade



Meaning:

A ten-fold change in $Q = [\text{Fe}^{2+}]/[\text{Fe}^{3+}]$ changes the cell voltage by 59 mV.

A probe to monitor Ferrous/Ferric

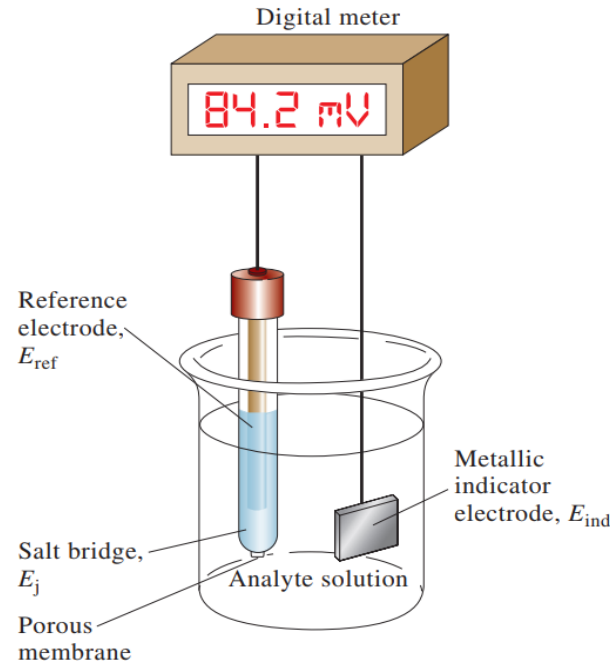
Note: we commonly plot a semi-log graph to show the calibration function.

Here we will assume that the activity of the solution is equal to the concentration (by way of the constant)

Recap: Components and Cell Potential Equation

Every potentiometric measurement consists of three potential contributions:

By convention, the **reference electrode** is always treated as the left-hand electrode in potentiometric measurements.



The **indicator (working) electrode** is immersed in the analyte solution and develops a potential that depends on the activity of the target ion.

The **salt bridge (or junction)** connects the two half-cells, allowing charge balance while minimizing ion mixing. It introduces a small liquid junction potential that cannot be fully eliminated.

2. Indicator (Working) Electrodes

An ideal indicator electrode:

- Responds rapidly and reproducibly to changes in analyte activity
- Exhibits a Nernstian potential–activity (concentration) relationship
- Provides reasonable selectivity toward the target ion or species

Main Types of Indicator Electrodes:

• **Metallic Indicator Electrodes**

- *First kind*: metal in equilibrium with its ions (e.g., $\text{Cu} \mid \text{Cu}^{2+}$)
- *Second kind*: metal coated with a slightly soluble salt (e.g., $\text{Ag} \mid \text{AgCl}$)
- *Inert redox electrodes*: Pt or Au sensing redox couples

• **Membrane Electrodes (Ion-Selective Electrodes, ISEs)**

- Include glass pH electrodes and polymer/plastic membranes
- Selectively respond to specific ions (F^- , Ca^{2+} , K^+ , etc.)

• **Gas-Sensing Probes**

- Gas-permeable membrane + internal solution + indicator electrode
- Used for CO_2 , NH_3 , and similar species that convert to ions after diffusion

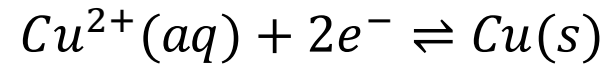
• **ISFET Electrodes**

- Solid-state, field-effect sensors
- Fast, miniaturized pH and ion detection

Electrodes of the First Kind

A first-kind indicator electrode is a **pure metal in direct equilibrium with its metal ion** in solution.

Example: the equilibrium between a copper and its cation Cu^{2+} :

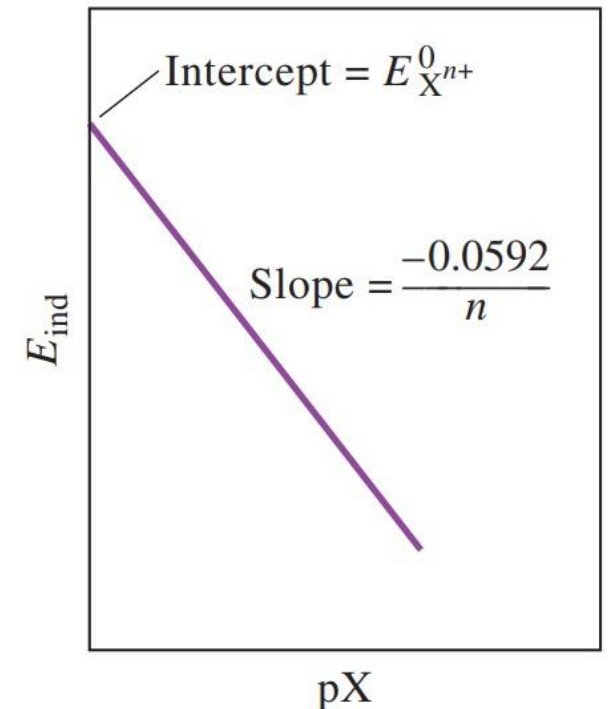


For such electrode the Nernst Equation (for a metal M) is:

$$E_{\text{ind}} = E_{M^{n+}/M}^{\circ} + \frac{0.0592}{n} \log a_{M^{n+}}$$

or using pX (where $\text{pX} = -\log a_{(M^{n+})}$): $E_{\text{ind}} = E_{M^{n+}/M}^{\circ} - \frac{0.0592}{n} \text{pX}$

Expected slope (at 25°C) : $-\frac{0.0592}{n} \text{ V per decade}$



$a_{M^{n+}}$: the activity of the ion (or in dilute solution, approximately its molar concentration, $[\text{Cu}^{2+}]$)

Limitations of First-Kind Electrodes

Why they are rarely used in potentiometry

- **Poor selectivity:** Respond to other easily reduced metal ions.
(Example: Cu electrode also responds to Ag^+ → cannot determine Cu^{2+} accurately.)
- **Restrictions based on pH:** Many metals dissolve in acids or bases → limited working range.
- **Oxygen effects:** Some metals (e.g., Fe, Cr, Co, Ni) oxidize in air → unreliable potentials.
Must deaerate (Bubble N_2 or Ar gas) solutions → inconvenient.
- **Irregular slopes:** Experimental slopes often deviate significantly from the ideal $-0.0592/n$

Where they are used

- Only a few first-kind systems are reliable:
- Ag/Ag^+ , Hg/Hg^{2+}
- Cu/Cu^{2+} , Zn/Zn^{2+} , Cd/Cd^{2+}
- Bi^{3+}/Bi , Tl^+/Tl , Pb^{2+}/Pb (in deaerated solutions)

Electrodes of the Second Kind

A second-kind electrode is a **metal coated with a sparingly soluble salt of that same metal**. Its potential responds **to the anion associated with that salt**.

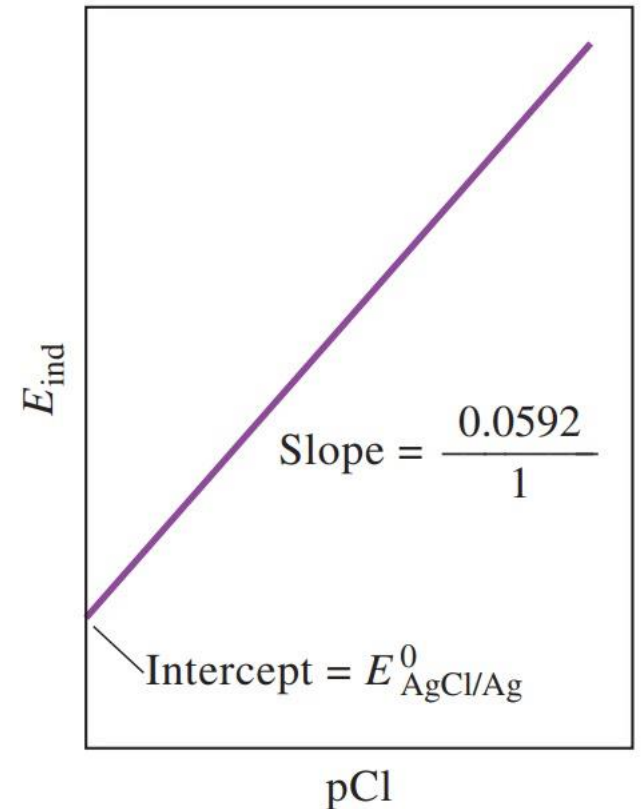
Example:

Nernst Equation
$$E_{\text{ind}} = E_{\text{AgCl/Ag}}^{\circ} - \frac{0.0592}{1} \log a_{\text{Cl}^-}$$

or using $p\text{Cl} = -\log a_{\text{Cl}^-}$:
$$E_{\text{ind}} = E_{\text{AgCl/Ag}}^{\circ} + 0.0592 p\text{Cl}$$

Key Features

- Responds to anion, not the metal ion
- Positive slope **+59 mV** per decade (for $n=1$, at 25°C)
- Very reproducible and widely used
(Ag/AgCl reference electrode is the classic example)

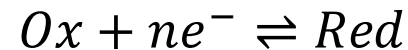


Inert Metallic Electrodes

- Made of chemically inert conductors such as Pt, Au, Pd, or carbon
- Do not form their own ions or sparingly soluble salts
- Provide only a surface for electron transfer
- Used when the analyte is a redox couple in solution (no metal electrode possible)

How they work

The metal surface establishes equilibrium with a solution redox pair, for example:



The potential depends solely on the activities of the oxidized and reduced forms:

$$E_{\text{ind}} = E^0 + \frac{0.0592}{n} \log \frac{a_{Ox}}{a_{Red}} \quad \text{(using Ox/Red form} \rightarrow \text{positive sign)}$$

There is no metal ion in the Nernst equation: the electrode is just an electron conduit.

Inert Metallic Electrodes

Unlike first- and second-kind electrodes, inert metallic electrodes **do not have a fixed slope sign**. The sign depends on how the redox ratio is written:

$$\text{If x-axis} = \log(\text{Ox/Red}) \rightarrow + \frac{0.0592}{n}$$

$$\text{If x-axis} = \log(\text{Red/Ox}) \rightarrow - \frac{0.0592}{n}$$

(using Ox/Red form $\rightarrow$ positive sign)

In all cases, the magnitude of the slope is $0.0592/n$ V per decade at 25 °C.

In the $\text{Fe}^{3+}/\text{Fe}^{2+}$ (Pt) example we defined: $Q = \frac{[\text{Fe}^{2+}]}{[\text{Fe}^{3+}]}$ So: $E = E^0 - 0.0592 \log Q$

This gave a -59 mV/decade slope in the calibration plot.

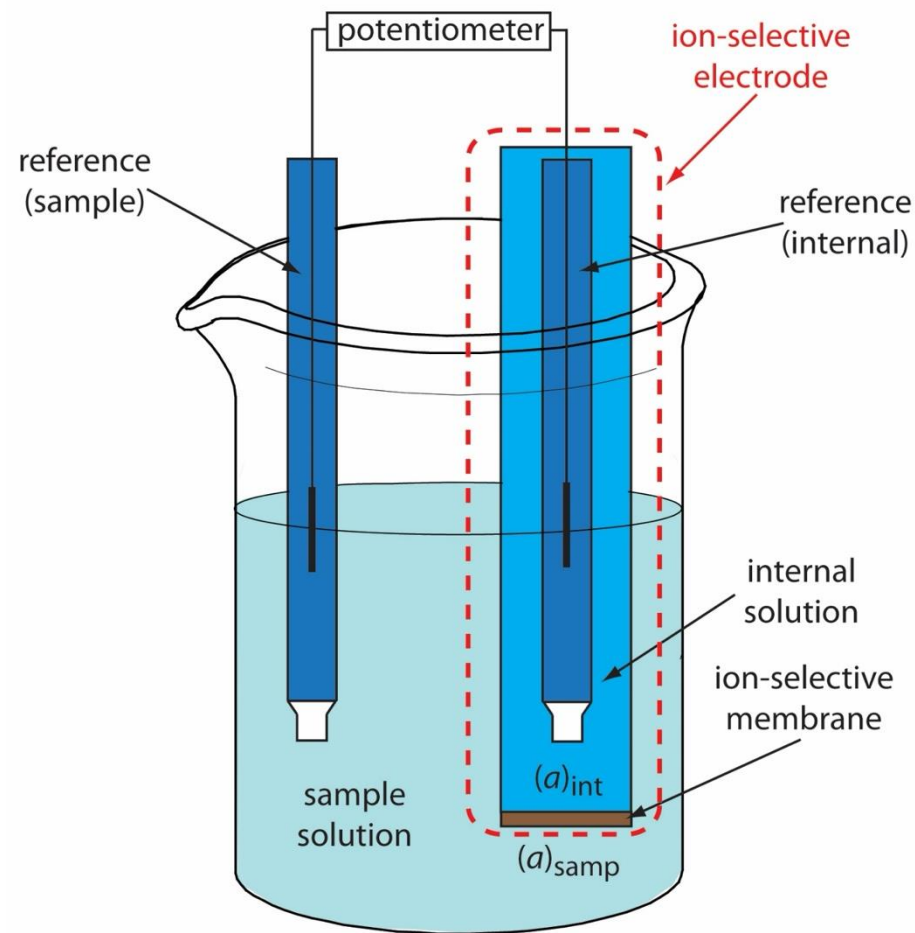
The slope was negative because of the definition of Q , not because Pt behaves like a “negative-slope electrode.

Where inert electrodes are used

- Redox titrations ($\text{Fe}^{2+}/\text{Fe}^{3+}$, $\text{Ce}^{3+}/\text{Ce}^{4+}$, $\text{Cr}^{2+}/\text{Cr}^{3+}$)
- Monitoring dissolved oxygen (Pt/ O_2)
- Any system with no suitable metal-metal ion equilibrium

Ion Selective Electrodes (ISE)

Membrane electrodes are sometimes called p-ion electrodes because the data obtained from them are usually presented as p-functions, such as pH, pCa, or pNO₃.



Ion-selective electrodes are used to monitor many different chemical species.

No redox reaction occurs at the membrane (the potential arises from ion-selective partitioning).

A thin membrane is configured to bind only a single ion of interest.

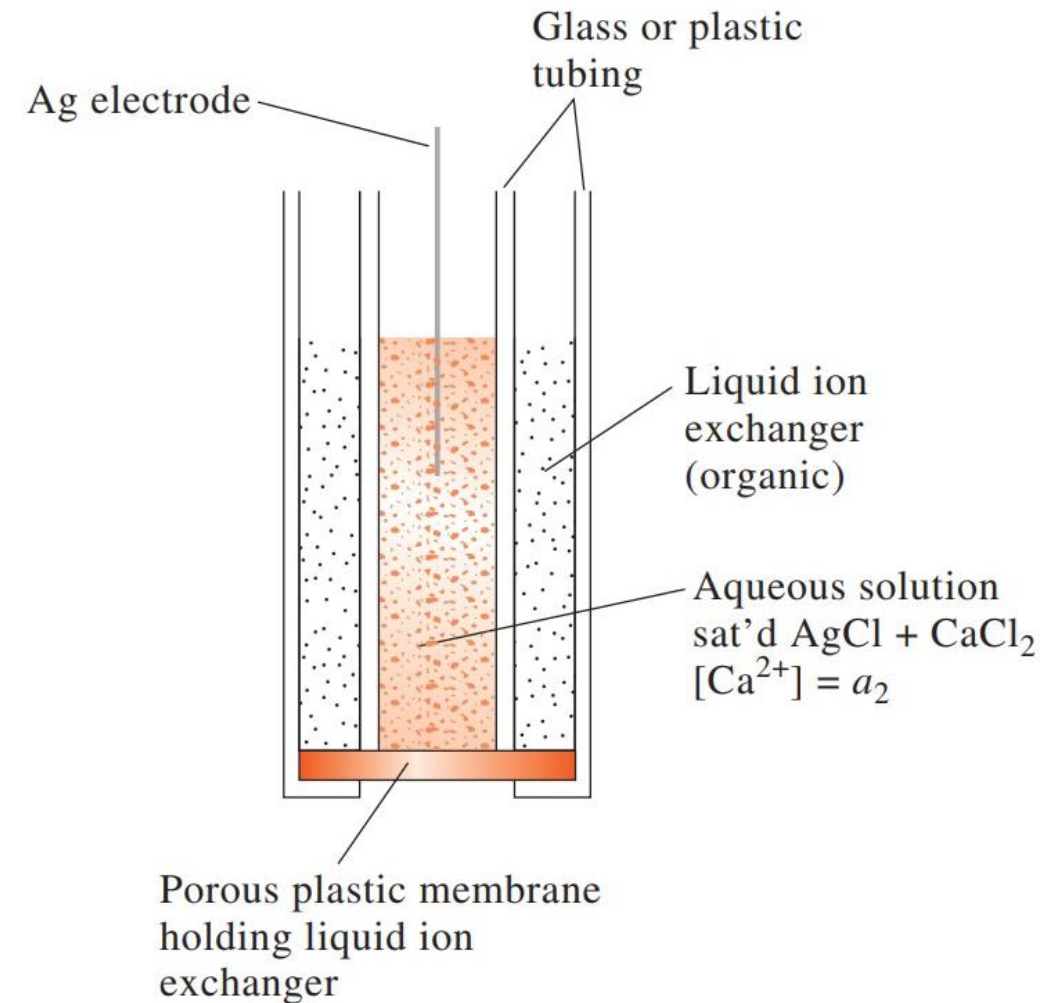
Two reference electrodes measure the potential difference across the membrane.

There are many physical constructions and chemical mechanisms, but only a few types will be examined:

- Liquid-Membrane Electrodes
- Crystalline-Membrane Electrodes
- The Glass Electrode for Measuring pH
- Glass Electrodes for Other Cations

Liquid-Membrane Electrodes

Developed for direct potentiometric measurement of **polyvalent cations** and certain anions.



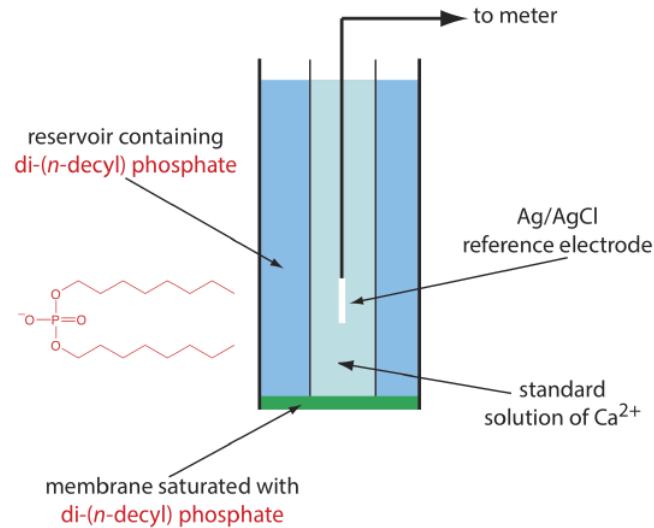
Basic Structure

- A conducting **membrane** that selectively binds a specific ion (e.g., Ca²⁺).
- An internal solution containing a fixed concentration of that ion.
- A silver electrode coated with AgCl serving as the internal reference.

The potential develops at the interface between:

- the solution containing the analyte, and
- a **liquid ion exchanger** that selectively binds the analyte ion.

Liquid-Membrane Electrodes



- Uses a porous plastic membrane saturated with the cation exchanger di-(n-decyl) phosphate
- Outer reservoir: di-(n-decyl) phosphate dissolved in an immiscible organic phase (e.g., di-n-octylphenylphosphonate); this solution saturates the porous membrane
- Inner reservoir: aqueous standard Ca^{2+} solution (a_2) and an Ag/AgCl reference electrode

Ca^{2+} -Selective Liquid-Membrane Electrode: $\text{Ca}^{2+} (aq) + 2 (\text{C}_{10}\text{H}_{21}\text{O})_2\text{PO}^- (mem) \rightleftharpoons \text{Ca}[(\text{C}_{10}\text{H}_{21}\text{O})_2\text{PO}_2]_2 (mem)$

A potential develops across the membrane when the extent of ion-exchanger dissociation differs on the two sides.

$$E_{\text{membrane}} = E_1 - E_2 = \frac{0.0592}{2} \log \frac{a_1}{a_2}$$

The membrane potential depends on the ratio of Ca^{2+} activity outside (a_1) vs inside (a_2). Since a_2 is constant, the potential becomes a linear function of pCa.

$$E_{\text{membrane}} = N + \frac{0.0592}{2} \log a_1 = N - \frac{0.0592}{2} \text{pCa}$$

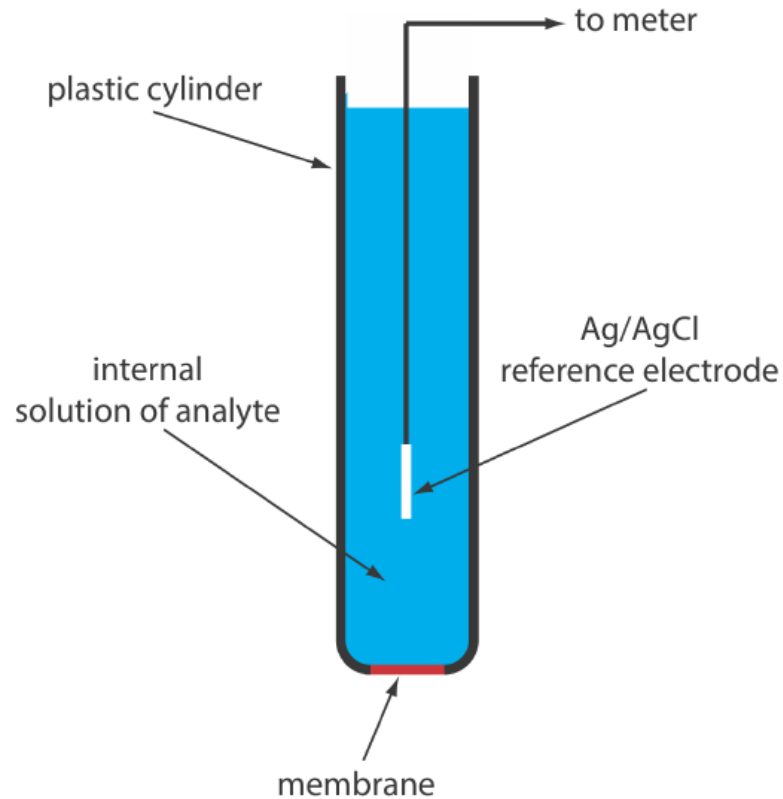
Liquid-Membrane Electrodes



Characteristics of Liquid-Membrane Electrodes*

Analyte Ion	Concentration Range, M [†]	Major Interferences [‡]
NH ₄ ⁺	10 ⁰ to 5 × 10 ⁻⁷	<1 H ⁺ , 5 × 10 ⁻¹ Li ⁺ , 8 × 10 ⁻² , Na ⁺ , 6 × 10 ⁻⁴ K ⁺ , 5 × 10 ⁻² Cs ⁺ , >1 Mg ²⁺ , >1 Ca ²⁺ , >1 Sr ²⁺ , >0.5 Sr ²⁺ , 1 × 10 ⁻² Zn ²⁺
Cd ²⁺	10 ⁰ to 5 × 10 ⁻⁷	Hg ²⁺ and Ag ⁺ (poisons electrode at >10 ⁻⁷ M), Fe ³⁺ (at >0.1[Cd ²⁺]), Pb ²⁺ (at >[Cd ²⁺]), Cu ²⁺ (possible)
Ca ²⁺	10 ⁰ to 5 × 10 ⁻⁷	10 ⁻⁵ Pb ²⁺ ; 4 × 10 ⁻³ Hg ²⁺ , H ⁺ , 6 × 10 ⁻³ Sr ²⁺ ; 2 × 10 ⁻² Fe ²⁺ ; 4 × 10 ⁻² Cu ²⁺ ; 5 × 10 ⁻² Ni ²⁺ ; 0.2 NH ₃ ; 0.2 Na ⁺ ; 0.3 Tris ⁺ ; 0.3 Li ⁺ ; 0.4 K ⁺ ; 0.7 Ba ²⁺ ; 1.0 Zn ²⁺ ; 1.0 Mg ²⁺
Cl ⁻	10 ⁰ to 5 × 10 ⁻⁶	Maximum allowable ratio of interferent to [Cl ⁻]: OH ⁻ 80, Br ⁻ 3 × 10 ⁻³ , I ⁻ 5 × 10 ⁻⁷ , S ²⁻ 10 ⁻⁶ , CN ⁻ 2 × 10 ⁻⁷ , NH ₃ 0.12, S ₂ O ₃ ²⁻ 0.01
BF ₄ ⁻	10 ⁰ to 7 × 10 ⁻⁶	5 × 10 ⁻⁷ ClO ₄ ⁻ ; 5 × 10 ⁻⁶ I ⁻ ; 5 × 10 ⁻⁵ ClO ₃ ⁻ ; 5 × 10 ⁻⁴ CN ⁻ ; 10 ⁻³ Br ⁻ ; 10 ⁻³ NO ₂ ⁻ ; 5 × 10 ⁻³ NO ₃ ⁻ ; 3 × 10 ⁻³ HCO ₃ ⁻ ; 5 × 10 ⁻² Cl ⁻ ; 8 × 10 ⁻² H ₂ PO ₄ ⁻ , HPO ₄ ²⁻ , PO ₄ ³⁻ ; 0.2 OAc ⁻ ; 0.6 F ⁻ ; 1.0 SO ₄ ²⁻
NO ₃ ⁻	10 ⁰ to 7 × 10 ⁻⁶	10 ⁻⁷ ClO ₄ ⁻ ; 5 × 10 ⁻⁶ ; I ⁻ ; 5 × 10 ⁻⁵ ClO ₃ ⁻ ; 10 ⁻⁴ CN ⁻ ; 7 × 10 ⁻⁴ Br ⁻ ; 10 ⁻³ HS ⁻ ; 10 ⁻² HCO ₃ ⁻ ; 2 × 10 ⁻² CO ₃ ²⁻ ; 3 × 10 ⁻² Cl ⁻ ; 5 × 10 ⁻² H ₂ PO ₄ ⁻ , HPO ₄ ²⁻ , PO ₄ ³⁻ ; 0.2 OAc ⁻ ; 0.6 F ⁻ ; 1.0 SO ₄ ²⁻
NO ₂ ⁻	1.4 × 10 ⁻⁶ to 3.6 × 10 ⁻⁶	7 × 10 ⁻¹ salicylate, 2 × 10 ⁻³ I ⁻ , 10 ⁻¹ Br ⁻ , 3 × 10 ⁻¹ ClO ₃ ⁻ , 2 × 10 ⁻¹ acetate, 2 × 10 ⁻¹ HCO ₃ ⁻ , 2 × 10 ⁻¹ NO ₃ ⁻ , 2 × 10 ⁻¹ SO ₄ ²⁻ , 1 × 10 ⁻¹ Cl ⁻ , 1 × 10 ⁻¹ ClO ₄ ⁻ , 1 × 10 ⁻¹ F ⁻
ClO ₄ ⁻	10 ⁰ to 7 × 10 ⁻⁶	2 × 10 ⁻³ I ⁻ ; 2 × 10 ⁻² ClO ₃ ⁻ ; 4 × 10 ⁻² CN ⁻ , Br ⁻ ; 5 × 10 ⁻² NO ₂ ⁻ , NO ₃ ⁻ ; 2 HCO ₃ ⁻ , CO ₃ ²⁻ ; Cl ⁻ , H ₂ PO ₄ ⁻ , HPO ₄ ²⁻ , PO ₄ ³⁻ , OAc ⁻ , F ⁻ , SO ₄ ²⁻
K ⁺	10 ⁰ to 1 × 10 ⁻⁶	3 × 10 ⁻⁴ Cs ⁺ ; 6 × 10 ⁻³ NH ₄ ⁺ , Tl ⁺ ; 10 ⁻² H ⁺ ; 1.0 Ag ⁺ , Tris ⁺ ; 2.0 Li ⁺ , Na ⁺
Water hardness (Ca ²⁺ + Mg ²⁺)	10 ⁻³ to 6 × 10 ⁻⁶	3 × 10 ⁻⁵ Cu ²⁺ , Zn ²⁺ ; 10 ⁻⁴ Ni ²⁺ ; 4 × 10 ⁻⁴ Sr ²⁺ ; 6 × 10 ⁻⁵ Fe ²⁺ ; 6 × 10 ⁻⁴ Ba ²⁺ ; 3 × 10 ⁻² Na ⁺ ; 0.1 K ⁺

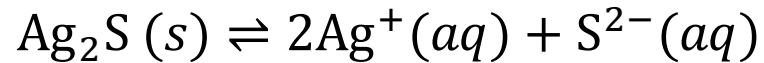
Crystalline-Membrane Electrodes (Solid-State ISEs)



- Ion-selective electrodes with membranes made from polycrystalline inorganic salts or single crystals.
- Prepared by sealing a 1–2 mm pellet (e.g., Ag_2S or mixtures of Ag_2S with other metal sulfides) into the end of a plastic cylinder.
- The cylinder is filled with an internal solution (fixed activity of target ion).
- An Ag/AgCl reference electrode is placed in the internal solution.
- The solid membrane makes contact with the sample solution.
- A potential develops due to differences in the extent of the solubility equilibrium at the membrane surface.

Crystalline-Membrane Electrodes (Solid-State ISEs)

Example: Silver Sulfide (Ag_2S) Membrane



- Responds to Ag^+ with:

$$E_{\text{cell}} = K + 0.0592 \log a_{\text{Ag}^+}$$

- Also responds to:

- S^{2-} :

$$E_{\text{cell}} = K - \frac{0.0592}{2} \log a_{\text{S}^{2-}}$$

- Cl^- : when AgCl is mixed with Ag_2S

$$E_{\text{cell}} = K - 0.0592 \log a_{\text{Cl}^-}$$

Mixed-Membrane Electrodes

- Ag_2S mixed with:

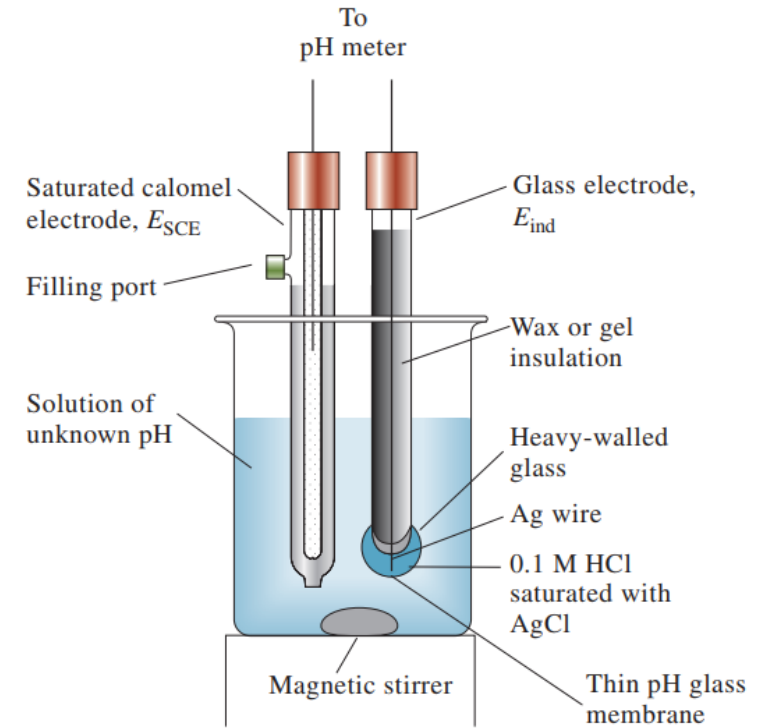
- $\text{CdS} \rightarrow$ selective for Cd^{2+}
- $\text{CuS} \rightarrow$ selective for Cu^{2+}
- $\text{PbS} \rightarrow$ selective for Pb^{2+}

$$E_{\text{cell}} = K + \frac{0.0592}{2} \log a_{\text{M}^{2+}}$$

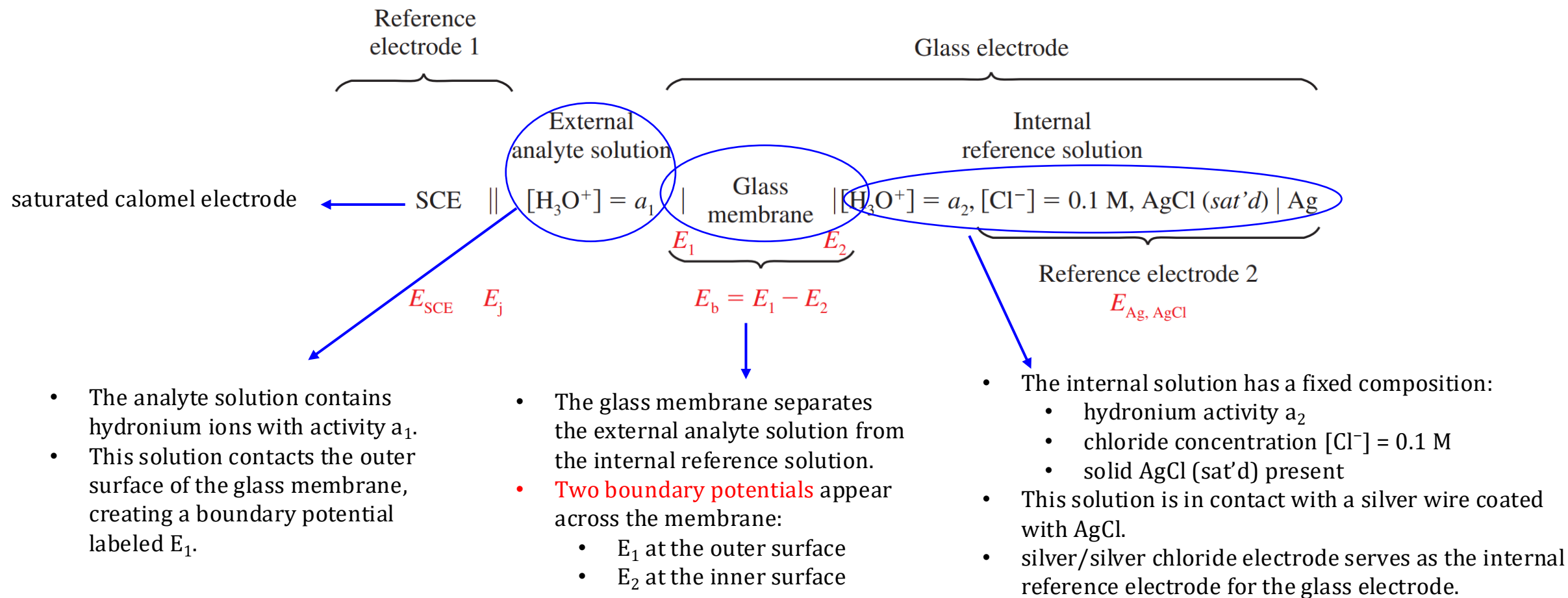


The Glass Electrode for Measuring pH

- A typical pH cell consists of a **glass indicator electrode (a thin pH-sensitive glass membrane)** and a saturated calomel reference electrode placed in the solution of unknown pH.
- Inside the tube is a small volume of dilute HCl saturated with AgCl.
- A silver wire in this solution forms an Ag/AgCl reference electrode, which is connected to the pH meter.
- The calomel electrode is connected to the other terminal of the measuring device.
- The glass electrode system therefore contains two reference electrodes:
 - the external calomel electrode
 - the internal Ag/AgCl electrode



The glass membrane (0.03–0.1 mm thick) has an electrical resistance of 50 to 500 M Ω .

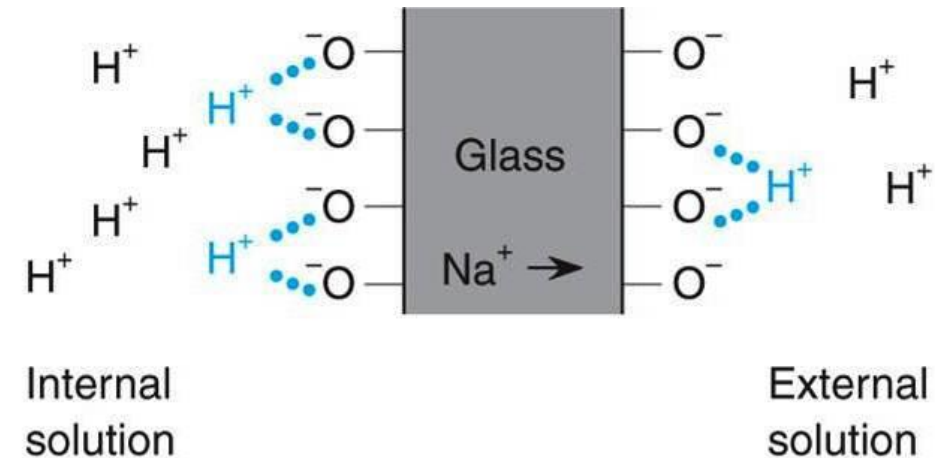


What Controls the Potential?

Where the Measured Potential Comes From

- The activity of hydrogen ions in the analyte solution establishes **an interfacial equilibrium that gives rise to the potential at the outer** glass surface.
- The **difference in H^+ activity across the membrane** produces the potential difference measured by the pH meter. This leads to a Nernst-type relationship:

$$E \propto \log \left(\frac{a_{out}}{a_{in}} \right)$$

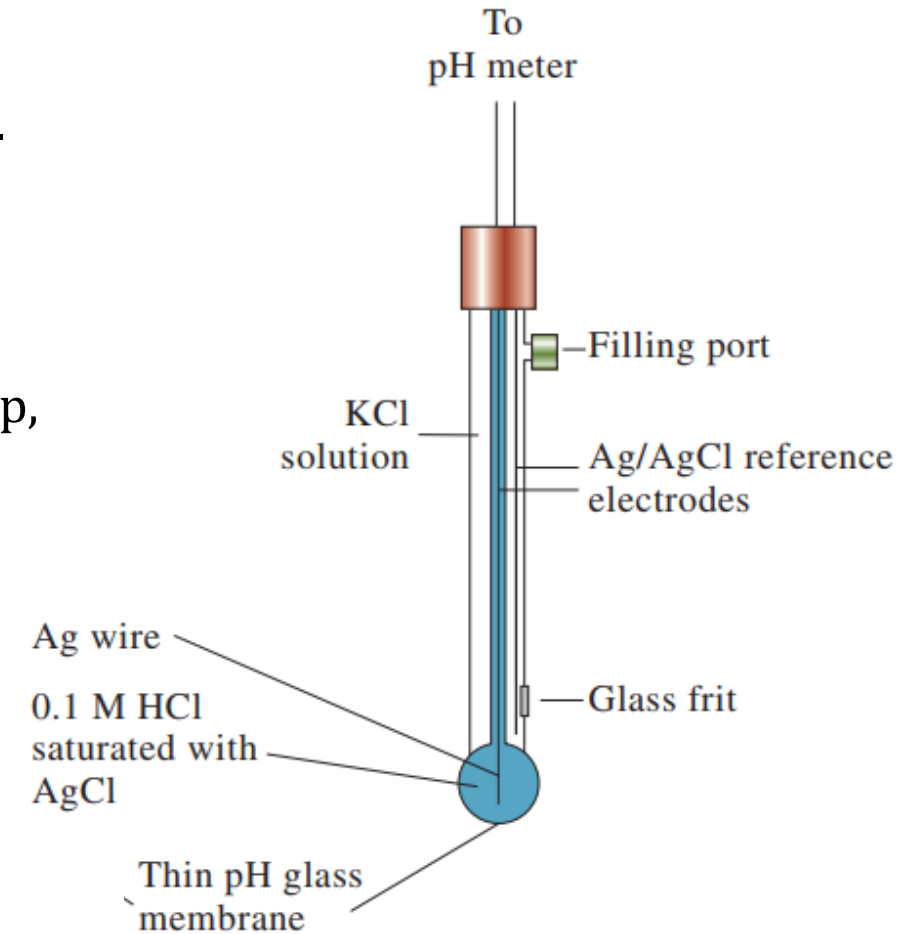
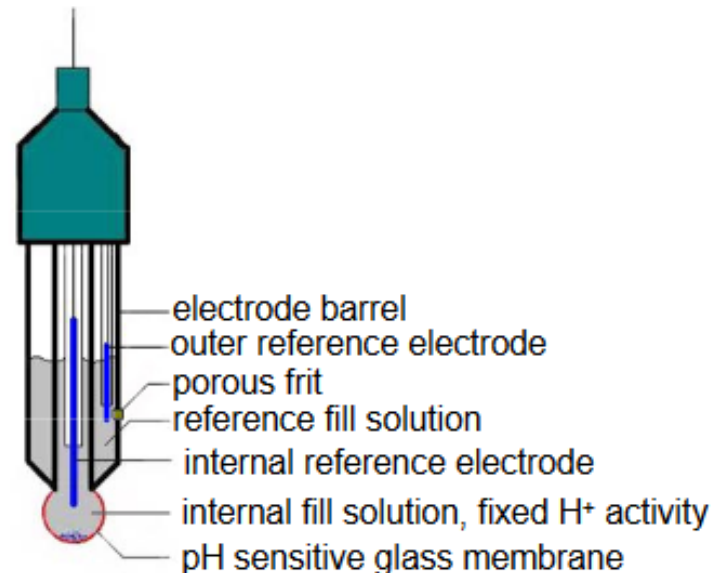


Function of the Two Reference Electrodes

- The internal and external reference electrodes only provide electrical contact to each side of the glass membrane.
- Their potentials are essentially constant, except for the junction potential, which changes slightly with the analyte composition.
- The potentials of the reference electrodes are set by the redox couples they contain.

Single-Probe Glass Electrodes

- In a typical pH meter the glass electrode and its Ag/AgCl internal reference electrode are placed at the center of a cylindrical probe.
- Surrounding them is ...the external reference electrode (typically Ag/AgCl in KCl filling solution).
- The external reference is less obvious than in the dual-probe setup, but the combination design is more convenient and can be made much smaller.

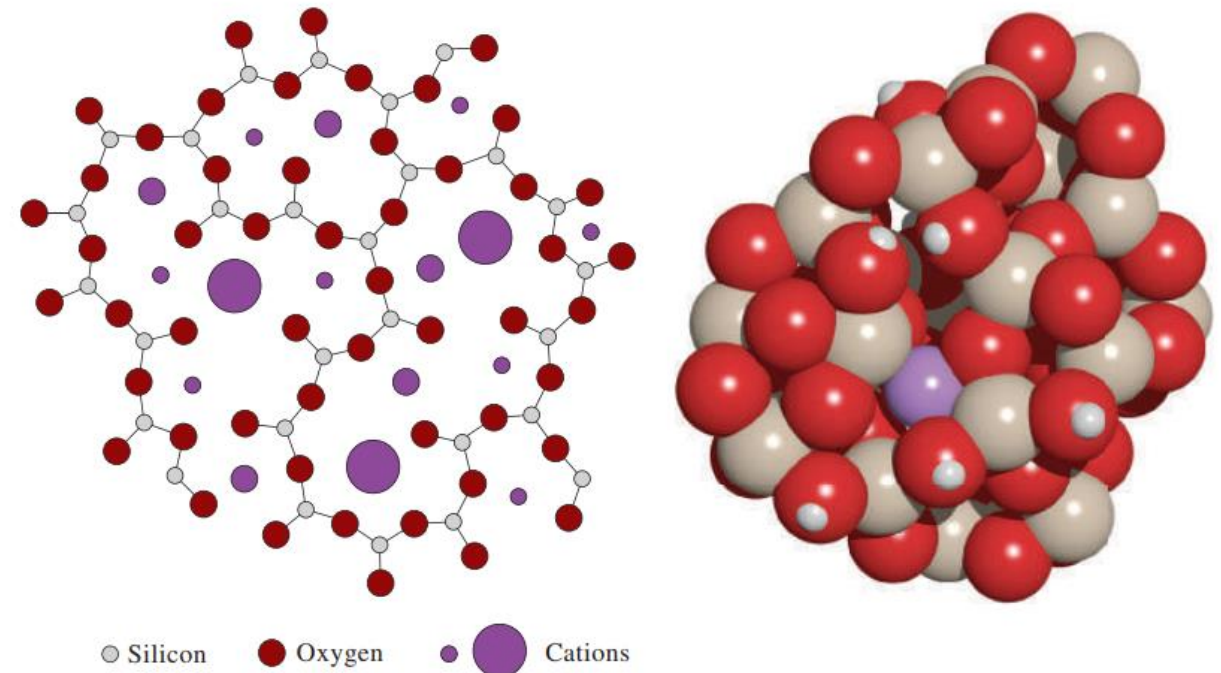


Glass Composition Matters

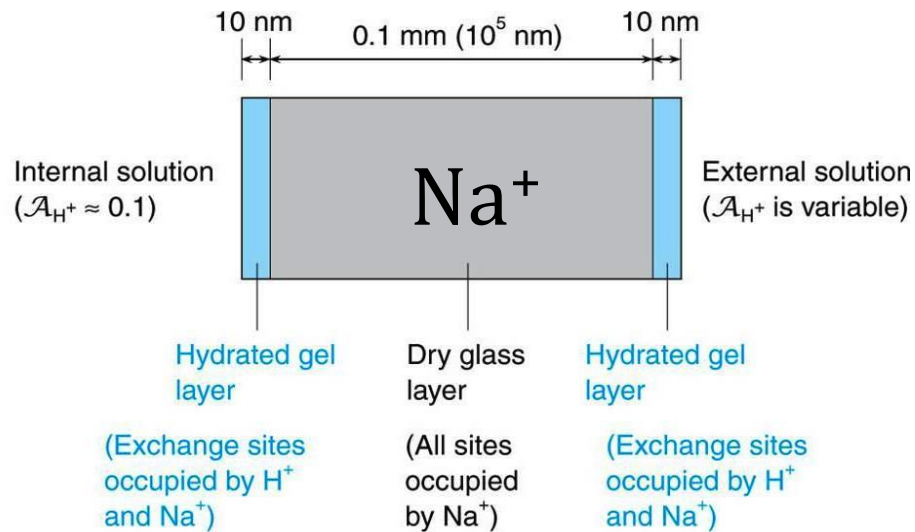
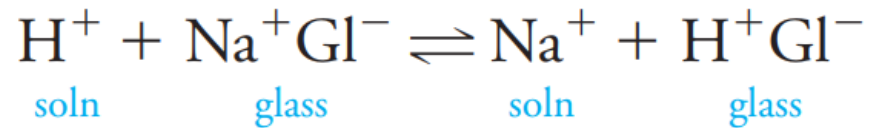
- Glass membranes used in pH electrodes are made from special silicate glasses.
- Corning 015 glass (widely used) contains: 22% Na_2O , 6% CaO , 72% SiO_2
- These membranes show excellent specificity to H^+ up to $\text{pH} \approx 9$.
- **At higher pH**, the membrane becomes somewhat responsive to Na^+ .
- Other glass formulations replace $\text{Na}^+/\text{Ca}^{2+}$ with Ba^{2+} or Li^+ to improve selectivity and lifetime.

Hydration Is Required

- Both surfaces of the glass membrane must be **hydrated** before the electrode works.
- Non-hygroscopic glasses **show no pH response**.
- Even hygroscopic glasses lose sensitivity after dehydration (e.g., storage over desiccant), but regain it when soaked in water.



Boundary Potentials Across the Glass Membrane



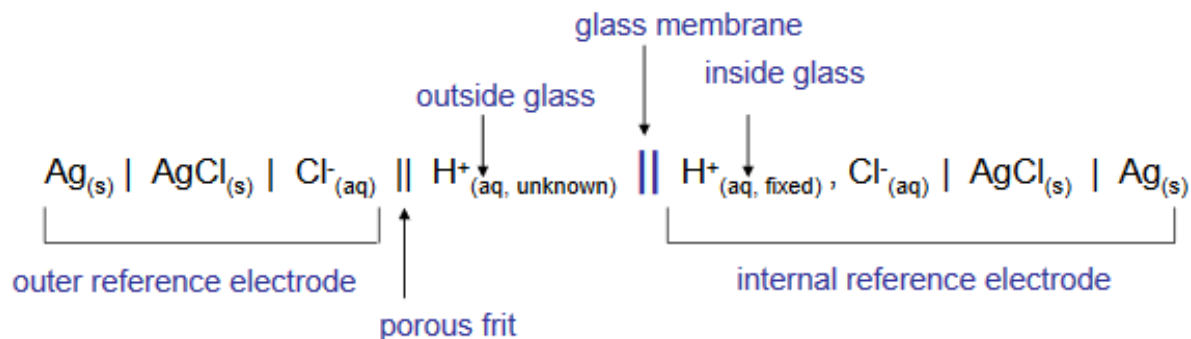
$$E_{\text{membrane}} = E_1 - E_2 = 0.0592 \log \frac{a_1}{a_2}$$

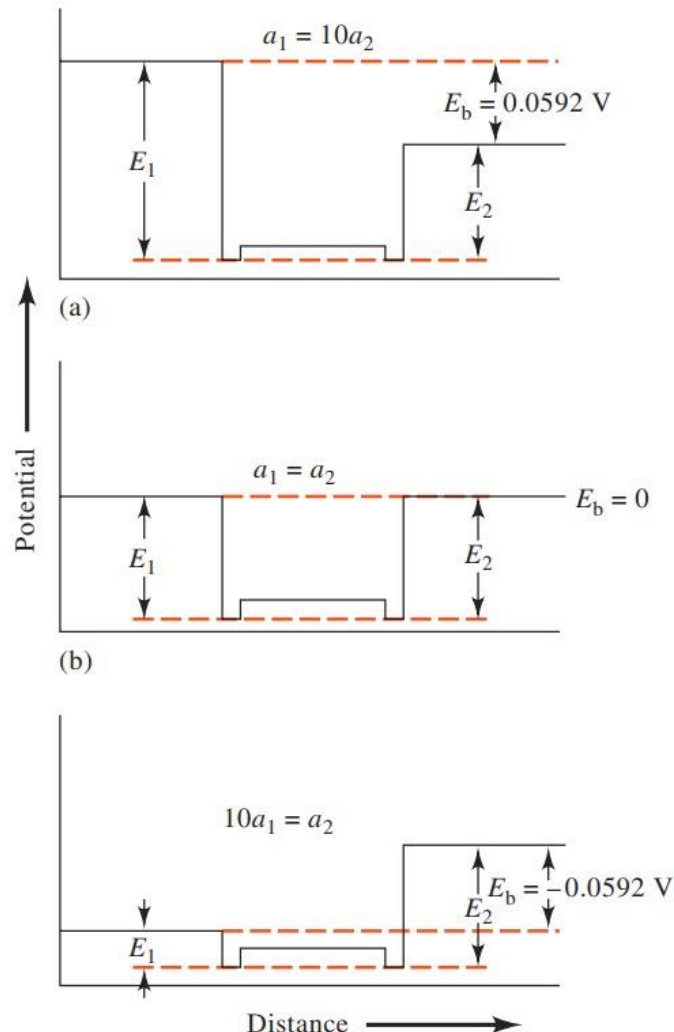
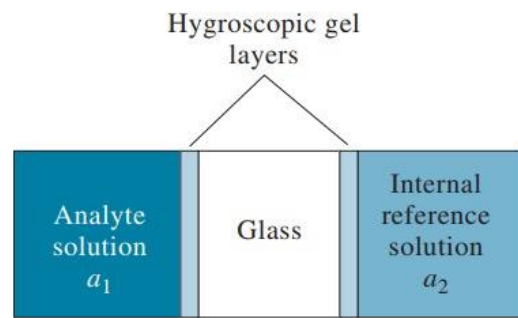
The hydrogen ion activity of the internal solution, a_2 , is constant ($L' = -0.0592 \log a_2$)

$$E_{\text{membrane}} = \text{constant} + 0.0592 \log a_1$$

Or

$$E_{\text{membrane}} = \text{constant} - 0.0592 \log pH$$





$$E_{\text{membrane}} = E_1 - E_2 = 0.0592 \log \frac{a_1}{a_2}$$

The hydrogen ion activity of the internal solution, a_2 , is constant ($L' = -0.0592 \log a_2$)

$$E_{\text{membrane}} = \text{constant} + 0.0592 \log a_1$$

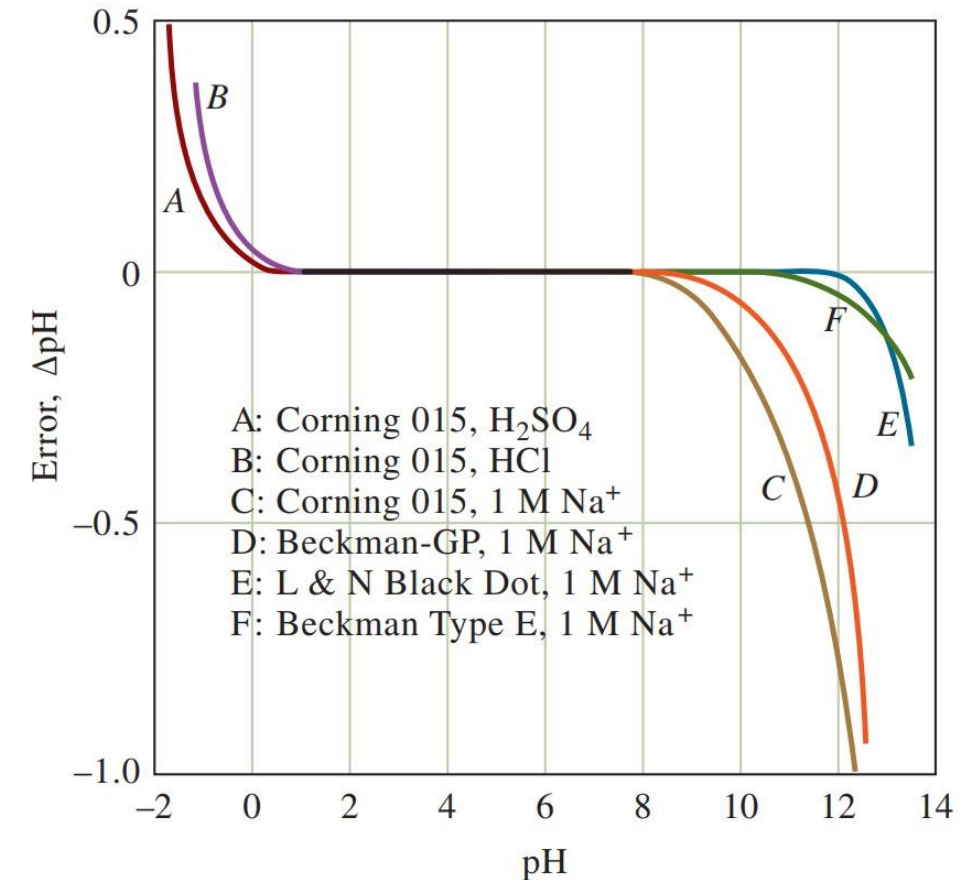
Or

$$E_{\text{membrane}} = \text{constant} - 0.0592 \log pH$$

Errors in Glass Electrode Measurements

Acid Error (Low-pH Error)

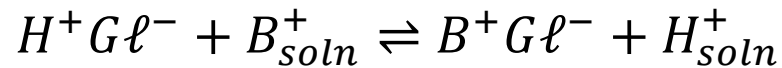
- Occurs at around $pH \lesssim \sim 0.5 - 1.0$.
- Measured pH is too high ($pH_{read} - pH_{true} > 0$).
- Caused by surface-site saturation: all exchange sites on the glass become occupied by H^+ .
- When the surface is saturated, the electrode cannot respond to further increases in $[H^+]$, so the reading stops decreasing.
- Magnitude varies between electrodes and is often poorly reproducible.



Errors in Glass Electrode Measurements

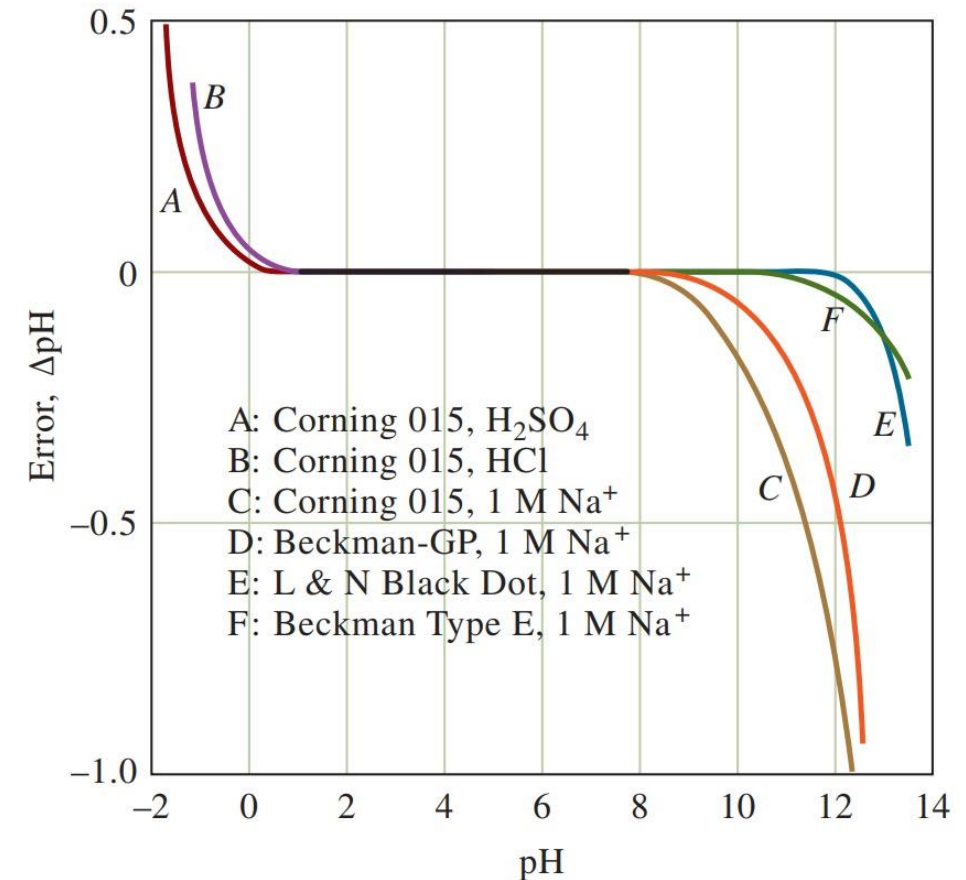
Alkaline Error (Sodium Error)

- Occurs at high pH, usually $pH \gtrsim 11 - 12$.
- Measured pH is too low ($pH_{read} - pH_{true} < 0$).
- Caused by the glass responding to alkali ions (mainly Na^+) as well as H^+ .
- Ion-exchange equilibrium on the glass surface:



where $B^+ = Na^+, K^+$, etc.

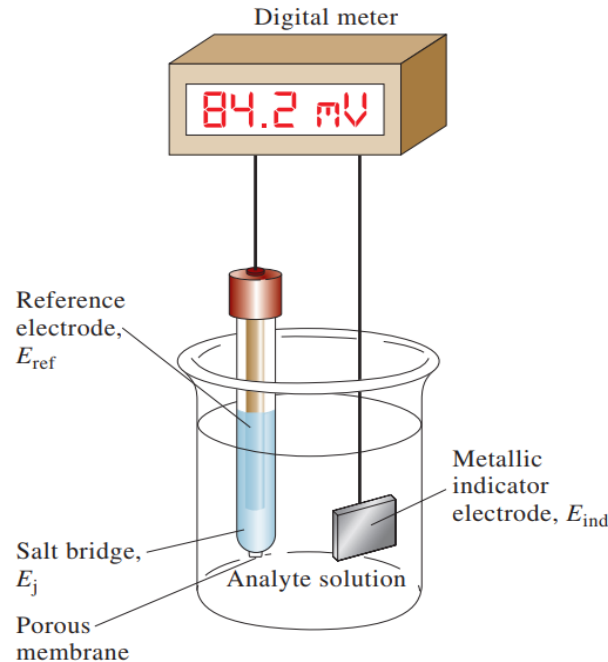
- The effect increases when $[Na^+]$ is high and $[H^+]$ is very low, making the electrode respond partly to B^+ instead of H^+ .



Recap: Components and Cell Potential Equation

Every potentiometric measurement consists of three potential contributions:

By convention, the **reference electrode** is always treated as the left-hand electrode in potentiometric measurements.



The **indicator (working) electrode** is immersed in the analyte solution and develops a potential that depends on the activity of the target ion.

The **salt bridge (or junction)** connects the two half-cells, allowing charge balance while minimizing ion mixing. It introduces a small liquid junction potential that cannot be fully eliminated.

3. Liquid Junction Potentials (E_j)

E_j develops at the **interface between two electrolyte** solutions of different composition or concentration, as a result of an unequal distribution of cations and anions across the boundary.

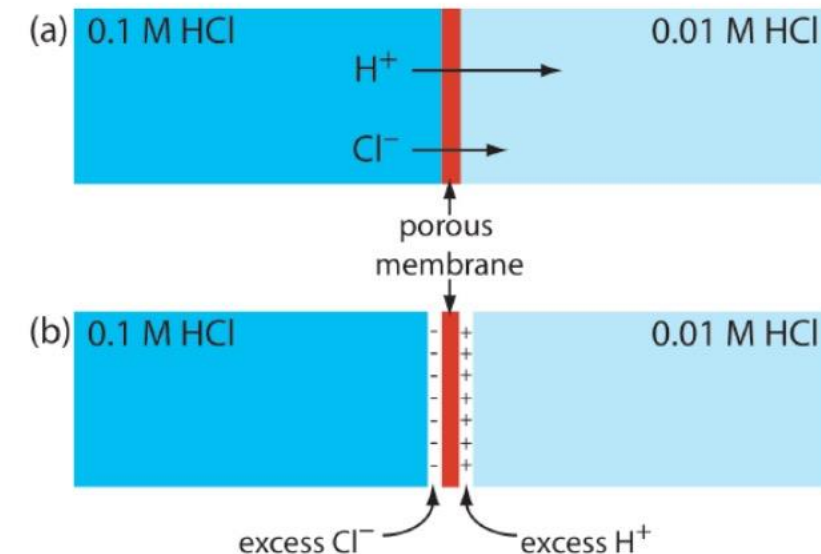
Ions **diffuse at different rates** when solutions of different concentrations meet.

Example: **At the interface of 0.1 M HCl | 0.01 M HCl**

- Fast H^+ diffusion into the dilute side causes it to accumulate excess positive charge.
- The concentrated side is left with excess Cl^- and becomes negatively charged.
- Charge separation continues until an electrostatic potential builds up that opposes further net diffusion.

This unequal mobility creates a charge separation, generating a measurable potential difference.

The side receiving more fast-moving ions becomes positively charged; the other side becomes negative $\rightarrow E_j$ forms.



Both H^+ and Cl^- diffuse from concentrated $\rightarrow$ dilute side.
 H^+ diffuses faster than Cl^- .

Liquid Junction Potentials (E_j)

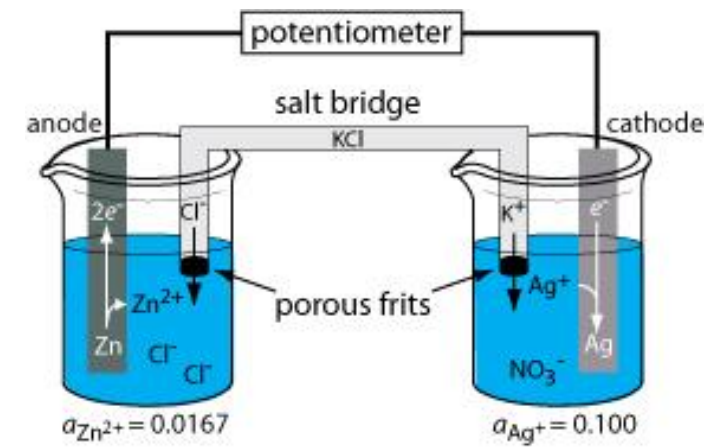
Magnitude: The junction potential is typically on the order of a few millivolts and can reach tens of millivolts for large concentration differences.

It depends on:

- ion concentrations on each side,
- ion mobilities.

Salt Bridges Reduce E_j

- A salt bridge minimizes the junction potential by using a salt whose ions have similar mobilities.
- KCl is commonly used because K^+ and Cl^- have similar ionic mobilities (transport numbers), which minimizes the junction potential..
- A KCl salt bridge produces a junction potential of only a few millivolts.
- Even with a well-designed bridge, a small residual E_j always remains.



Potentiometry Measurement

Potentiometry Measurement: Direct Potentiometry

The simplest potentiometric measurement compares the **potential of an indicator electrode in the analyte solution** with its **potential in standard solutions of known analyte concentration**.

If the electrode responds selectively to the analyte, no separation steps are required.

For a *cation – selective electrode*:
$$E_{cell} = \text{Constant} - \frac{0.0592}{n} pA$$

As the concentration (activity) of cation A increases: $pA = -\log a_A$ decreases.

Therefore, **E_{cell} increases with increasing cation** concentration (Since pA decreases as concentration increases, the negative sign causes E_{cell} to increase).

Instrument Polarity Convention

- Because this trend is opposite to the familiar pH scale, instrument manufacturers typically reverse the meter leads for cation-sensitive electrodes (such as glass electrodes).
- This ensures that the meter reading increases as pA increases, which corresponds to decreasing analyte concentration.

Potentiometry Measurement: Direct Potentiometry

The simplest potentiometric measurement compares the **potential of an indicator electrode in the analyte solution** with its **potential in standard solutions of known analyte concentration**.

If the electrode responds selectively to the analyte, no separation steps are required.

For a *anion – selective* electrode:

$$E_{cell} = \text{Constan} - \frac{0.0592}{n} pA$$

As the anion concentration (activity) a_{A-} increases, $pA = -\log a_{A-}$ decreases

Because of the negative sign, a decrease in pA causes an increase in E_{cell} .

Therefore: Higher anion concentration $\rightarrow$ higher cell potential.

Instrument Convention

- For anion-sensitive electrodes, the normal polarity gives intuitive readings (higher meter reading (higher pA) $\rightarrow$ lower analyte concentration), so no lead reversal is required.
- Result: The meter reading increases when analyte concentration decreases (just like the pH scale).

The Electrode-Calibration Method

Why Calibration Is Needed

The constant value in the Nernst-type equation contains several terms that cannot be calculated: Junction potential, Asymmetry potential, Standard-state reference terms. Therefore C must be determined experimentally.

$$E_{cell} = constant - \frac{0.0592}{n} pA \quad \text{solve for } pA \rightarrow \quad pA = \frac{n (constant - E_{cell})}{0.0592}$$

How the Calibration Method Works

- Measure E_{cell} for one or more standard solutions with known activity (or concentration).
- Fit the data to obtain the slope and intercept (effective C).
- Assume C remains unchanged between standards and the unknown sample.
- Then solve for pA (or pX)

Important Notes

- Multiple standards reduce error (slope often deviates from the ideal Nernst value).
- Membrane electrodes drift → recalibrate if measurements extend for hours.

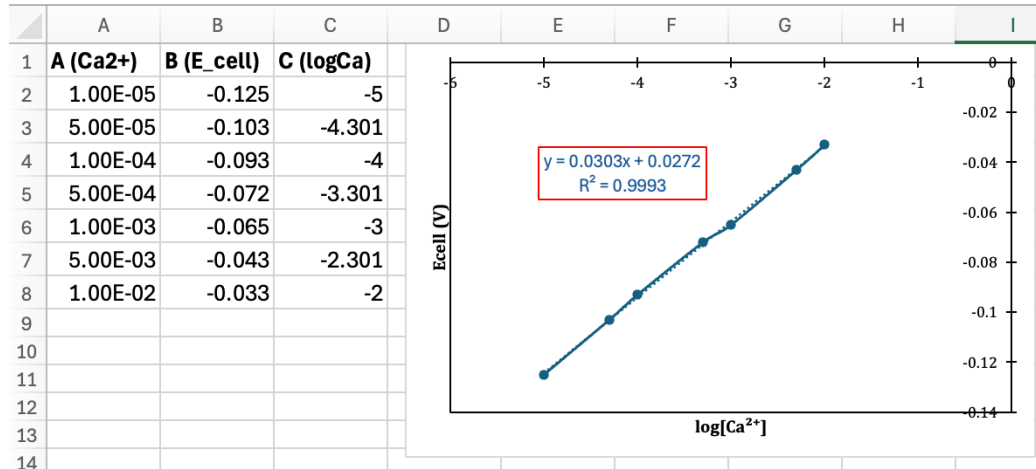
Practice Problem

A series of Ca^{2+} standards were prepared, each containing 0.5 M KNO_3 to maintain constant ionic strength. The measured cell potentials are shown below:

What is the concentration of Ca^{2+} in a water sample if its cell potential is -0.084 V?

(assuming activity $\approx$ concentration at constant ionic strength)

$[\text{Ca}^{2+}]$ (M)	E_{cell} (V)
1.00×10^{-5}	-0.125
5.00×10^{-5}	-0.103
1.00×10^{-4}	-0.093
5.00×10^{-4}	-0.072
1.00×10^{-3}	-0.065
5.00×10^{-3}	-0.043
1.00×10^{-2}	-0.033



Calibration equation from the graph is:

$$E_{\text{cell}} = 0.0303 \log[\text{Ca}^{2+}] + 0.0272$$

Substitute the sample E_{cell} into the calibration equation:

$$-0.084 = 0.0303 \log[\text{Ca}^{2+}] + 0.0272$$

$$\log[\text{Ca}^{2+}] = \frac{-0.084 - 0.0272}{0.0303} = -3.67$$

$$[\text{Ca}^{2+}] = 10^{\log[\text{Ca}^{2+}]} = 10^{-3.67} = 2.14 \times 10^{-4} \text{ M}$$

Note: The experimental slope (0.0303) is slightly higher than the ideal Nernst slope for Ca^{2+} ($0.05916/2 = 0.02958$).

This small deviation is normal for real ion-selective electrodes and is one reason why we calibrate using several standards instead of assuming the theoretical value.

The Standard Addition Method

Single-Point Standard Addition

1. Measure the potential of the original sample: $E'_{cell} = C + \frac{0.0592}{n} \log a_{\text{samp}} \rightarrow -\log a_{\text{samp}} = -\frac{E'_{cell} - C}{\frac{0.0592}{n}}$
2. Add a known volume of a standard solution, and measure the new potential:

$$E''_{cell} = C + \frac{0.0592}{n} \log a_{\text{after}} \rightarrow -\log a_{\text{after}} = -\frac{E''_{cell} - C}{\frac{0.0592}{n}}$$

Account for dilution of sample due to addition of the standard: $a_{\text{after}} = \frac{a_{\text{samp}} V_{\text{samp}} + a_{\text{std}} V_{\text{std}}}{V_{\text{samp}} + V_{\text{std}}}$

4. Eliminate the constant C:

$$-\log \left(\frac{a_{\text{after}}}{a_{\text{samp}}} \right) = \frac{n (E''_{cell} - E'_{cell})}{0.0592}$$

Important notes

- Assume the constant C remains unchanged between the two measurements.
- If $V_{\text{std}} \ll V_{\text{samp}}$, ionic strength and activity coefficients remain essentially constant.
- In practice, activity is often approximated by concentration.
- $-\log x = px$

Practice Problem

A cell made of a saturated calomel electrode and a lead-ion selective electrode gives a potential of -0.4706 V in 50.00 mL of a water sample. After adding 5.00 mL of 0.0200 M Pb^{2+} standard, the potential changes to -0.4490 V. Assume activity $\approx$ concentration, Calculate the molar concentration of Pb^{2+} in the original sample.

For Pb^{2+} ($n = 2$):

$$\text{pPb} = -\log[\text{Pb}^{2+}] = -\frac{E_{\text{cell}} - C}{\frac{0.0592}{2}}$$

Before addition:

$$-\log[\text{Pb}^{2+}] = -\frac{-0.4706 - C}{\frac{0.0592}{2}}$$

After addition:

$$[\text{Pb}^{2+}]_{\text{after}} = \frac{[\text{Pb}^{2+}] \times 50.00 + (0.0200) \times 5.00}{50.0 + 5.00} = 0.9091[\text{Pb}^{2+}] + 1.818 \times 10^{-3}$$

$$-\log(0.9091[\text{Pb}^{2+}] + 1.818 \times 10^{-3}) = -\frac{-0.4490 \text{ V} - C}{\frac{0.0592}{2}}$$

A cell made of a saturated calomel electrode and a lead-ion selective electrode gives a potential of -0.4706 V in 50.00 mL of a water sample. After adding 5.00 mL of 0.02000 M Pb^{2+} standard, the potential changes to -0.4490 V. Assume activity $\approx$ concentration, Calculate the molar concentration of Pb^{2+} in the original sample.

Subtract the two Nernst equations to eliminate C :

$$-\log \frac{[\text{Pb}^{2+}]}{0.9091[\text{Pb}^{2+}] + 1.818 \times 10^{-3}} = \frac{2 [-0.4490 - (-0.4706)]}{0.0592}$$

$$\rightarrow \frac{2 [-0.4490 - (-0.4706)]}{0.0592} = \frac{0.0432}{0.0592} = 0.7297$$

$$\text{So: } -\log \frac{[\text{Pb}^{2+}]}{0.9091[\text{Pb}^{2+}] + 1.818 \times 10^{-3}} = -0.7297 \quad \rightarrow \quad \frac{[\text{Pb}^{2+}]}{0.9091[\text{Pb}^{2+}] + 1.818 \times 10^{-3}} = 10^{-0.7297} = 0.1863$$

Now solve for $[\text{Pb}^{2+}]$:

$$[\text{Pb}^{2+}] = 0.1863 (0.9091[\text{Pb}^{2+}] + 1.818 \times 10^{-3}) \quad \rightarrow \quad [\text{Pb}^{2+}] = 0.1694[\text{Pb}^{2+}] + 3.388 \times 10^{-4}$$

$$0.8306[\text{Pb}^{2+}] = 3.388 \times 10^{-4}$$

$$\Rightarrow [\text{Pb}^{2+}] = 4.08 \times 10^{-4} \text{ M}$$

The Standard Addition Method

Multiple-Point Standard Addition

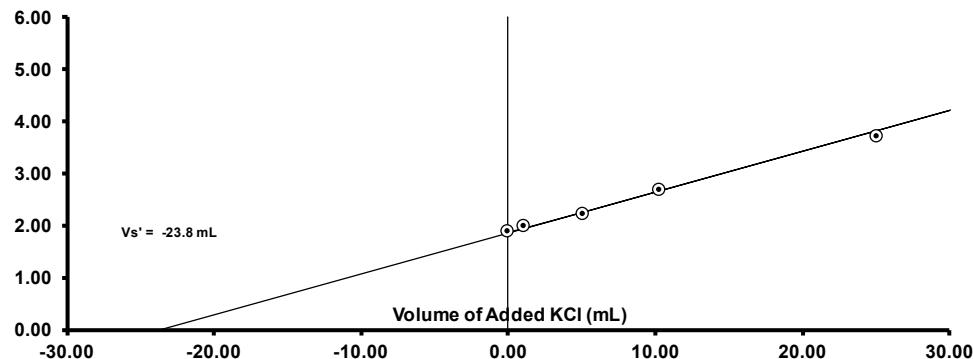
1. Measure the initial potential E_{samp} .
2. Add small, known volumes of standard (V_{std}) to the same beaker.
3. After each addition, measure the new potential $E_1, E_2, E_3, \dots$

For each addition, the analyte activity becomes: $a_{\text{after}} = \frac{a_{\text{samp}}V_{\text{samp}} + a_{\text{std}}V_{\text{std}}}{V_{\text{samp}} + V_{\text{std}}}$

The Nernst equation applies:

$$E = C + \frac{0.0592}{n} \log a$$

Standard Additions Plot for Chloride in Celery Salt
ISE Slope = -52.4 mV, [KCl] = 0.09986 M



- Because the constant C is the same for all additions, plotting E vs. added volume (V_{std}) gives a straight line.
- The x-intercept gives the *equivalent negative volume* needed to dilute the analyte to zero: $V_s' = -(\text{x-intercept})$
- Then the sample concentration is:

$$C_{\text{samp}} = C_{\text{std}} \left(\frac{V_s'}{V_{\text{samp}}} \right)$$

Thank you all for a wonderful semester.

**I truly hope you enjoyed the course as much
as I enjoyed teaching it this semester.**

**Please stay in touch, and best of luck on
your exams!**